Constrained Online Convex Optimization without Slater's Condition

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Abstract

We study constrained online convex optimization with adversarial losses and stochastic or adversarial constraints. For stochastic constraints, existing algorithms that achieve nearly optimal regret and constraint violation bounds typically rely on regularity assumptions such as Slater's condition, while adversarial-constraint algorithms avoid these assumptions by using a rather restrictive round-wise feasible comparator. We bridge this gap with an anytime primal-dual framework that incorporates an adaptive regularizer into the dual update. The regularizer stabilizes the dual process without relying on the negative drift induced by Slater's condition. For stochastic constraints and convex losses, our algorithm achieves $O(\sqrt{T})$ expected regret and $O(\sqrt{T}\log T)$ expected cumulative constraint violation. Furthermore, we show that our algorithm also admits high-probability bounds of the same order on regret and constraint violation. For strongly convex losses, the regret bound improves to $O(\log T)$ with a violation bound of the same order. With a minor modification, the framework also applies to adversarial constraints and provides guarantees for hard constraint violation.

1 Introduction

Constrained Online Convex Optimization (COCO) is a sequential decision-making problem under uncertainty in which, at each round, a decision maker selects an action without prior knowledge of the convex loss function, while feasibility requirements restrict the decision-making process. The goal is to minimize the cumulative loss while satisfying these requirements over time. COCO arises naturally in a wide variety of real-world applications, including online advertising with budget constraints (Mehta et al., 2007), portfolio optimization with risk constraints (Cover, 1991; Rockafellar et al., 2000), wireless network resource management under power constraints (Tse and Viswanath, 2005), and online recommendation systems with fairness constraints (Singh and Joachims, 2018).

The COCO problem is typically formulated as follows (a detailed formulation is provided in Section 2). In each round t, the decision maker selects an action $\mathbf{x}_t \in \mathcal{X}$. Then, the loss function f_t and the constraint function g_t are revealed to the decision maker, where feasibility is characterized by the functional constraint $g_t(\mathbf{x}) \leq 0$. Given a comparator $\mathbf{x}^\star \in \mathcal{X}$, the performance of the decision maker is measured by the regret and cumulative constraint violation over T rounds, denoted by $\mathrm{Regret}(T)$ and $\mathrm{Violation}(T)$, respectively:

$$\mathrm{Regret}(T) = \sum_{t=1}^{T}(f_t(\mathbf{x}_t) - f_t(\mathbf{x}^\star)), \quad \mathrm{Violation}(T) = \sum_{t=1}^{T} g_t(\mathbf{x}_t).$$

COCO has been studied under various settings. One line of work considers COCO with *stochastic constraints* (Yu et al., 2017; Wei et al., 2020). In this setting, the constraint functions are sampled i.i.d. from an underlying distribution, whereas the loss functions may vary arbitrarily over time. Yu et al. (2017) proposed a drift-plus-penalty algorithm that achieves $O(\sqrt{T})$ bounds on regret and cumulative constraint violation. However, their analysis relies on Slater's condition, which assumes the existence of a strictly feasible decision in expectation, i.e., there exist $\hat{\mathbf{x}}$ and $\varepsilon > 0$ such that $\mathbb{E}[g_1(\hat{\mathbf{x}})] = \cdots = \mathbb{E}[g_T(\hat{\mathbf{x}})] \leq -\varepsilon$, where the expectation is taken over the randomness of the constraint functions. More importantly, their bound on cumulative constraint violation is $O(\varepsilon^{-1}\sqrt{T})$, which deteriorates as ε approaches zero. This motivates us to investigate whether this condition can be relaxed while maintaining regret and constraint violation guarantees.

Wei et al. (2020) made some progress in this direction for the same setting by replacing Slater's condition with what they call the sequential existence of Lagrange multipliers (SELM) condition, which assumes the existence of uniformly bounded Lagrange multipliers. Since Slater's condition implies SELM, the latter is a weaker assumption. Nevertheless, their analysis still relies on the boundedness of the Lagrange multipliers, with the hidden constant in the constraint violation bound depending on the corresponding multiplier bound. Consequently, whether these guarantees can be achieved without such regularity assumptions has remained unresolved for COCO with stochastic constraints.

In contrast to the stochastic constraint setting, one can remove the necessity of a regularity assumption in another line of settings, called COCO with *adversarial constraints* against a *restrictive* comparator, where the constraint functions as well as the loss functions can vary arbitrarily over time (Yi et al., 2020; Sinha and Vaze, 2024). Here, the comparator is restrictive in that it is defined to be *round-wise feasible*: $g_t(\mathbf{x}^\star) \leq 0,\ \forall t = 1, \ldots, T$. For this setting, Yi et al. (2020) proposed an algorithm without assuming Slater's condition or bounded Lagrange multipliers, but its guarantees are suboptimal. Later, Sinha and Vaze (2024) proposed an algorithm that does not rely on these conditions but achieves nearly optimal regret and constraint violation bounds of $O(\sqrt{T})$ and $\widetilde{O}(\sqrt{T})$[1], respectively. The bounds are independent of any Slater constant or Lagrange multiplier bound. Additionally, Sarkar and Sinha (2026b) developed an anytime extension that does not require prior knowledge of the horizon T.

Given the success in the adversarial constraint setting, one may ask whether the same guarantees carry over to the stochastic constraint setting, since any realized stochastic

[1]Notation $\widetilde{O}(\cdot)$ is used to hide polylogarithmic factors, and the precise bound of Sinha and Vaze (2024) on constraint violation is $O(\sqrt{T}\log T)$.

constraint sequence can be viewed as a special case of adversarial constraint sequences. Unfortunately, the answer is no. The key distinction between the two settings lies in the definition of the comparator $\mathbf{x}^\star$, which plays a central role in the performance metrics. As mentioned above, the previous work for the adversarial setting takes a round-wise feasible comparator, but in the stochastic constraint setting, the comparator is defined to be *feasible in expectation*: $\mathbb{E}[g_1(\mathbf{x}^\star)] = \cdots = \mathbb{E}[g_T(\mathbf{x}^\star)] \leq 0$. This distinction implies that directly applying the algorithm of Sinha and Vaze (2024) to the stochastic constraint setting can provide guarantees only under the stronger assumption that $\mathbf{x}^\star$ is feasible almost surely, i.e., $g_t(\mathbf{x}^\star) \leq 0$ with probability 1. This yields a weaker notion of regret, which is not satisfactory for the stochastic constraint setting.

1.1 Our Contributions

Prior guarantees for COCO with stochastic constraints can be achieved only under additional assumptions or with respect to a weaker notion of regret. Motivated by this gap, this paper develops an algorithm for this setting that achieves nearly optimal guarantees[2] without these limitations. We summarize the main contributions below and provide a comparison with recent works in Table 1.

- We propose a novel algorithm (Algorithm 1) for COCO with stochastic constraints that does not rely on additional assumptions, including Slater's condition and bounded Lagrange multipliers. For the stochastic constraint setting, we show that our algorithm achieves expected regret and constraint violation bounds of $O(\sqrt{T})$ and $\widetilde{O}(\sqrt{T})$, respectively (Theorem 1). The novelty of our algorithm stems from incorporating an adaptive regularizer into the dual update. In principle, this step stabilizes the dual process without relying on the negative drift induced by Slater's condition. More technically, our regularizer is carefully designed to offset an accumulated term involving the derivative of a Lyapunov function that arises in the regret analysis.

- Beyond expected guarantees, we also provide high-probability guarantees. In particular, given a confidence parameter δ, with probability at least $1-\delta$, our algorithm achieves regret and violation bounds of order $O(\sqrt{T})$ and $\widetilde{O}(\sqrt{T})$, respectively (Theorem 2). The key step is to obtain an upper bound on the Lyapunov-weighted sum of comparator constraints by applying a concentration inequality for supermartingales. We then modify the regularizer appropriately so that it can offset this bound as well.

- We further extend the guarantees to strongly convex losses and adversarial constraints. Under stochastic constraints, when losses are strongly convex, our algorithm improves the expected and high-probability regret bounds to $O(\log T)$ and $O(\log T + \sqrt{\log(1/\delta)})$, respectively, while the constraint violation bounds remain the same as in the convex loss case (Theorems 3 and 4).

- Moreover, under adversarial constraints, a slightly modified version of our algorithm achieves regret and hard constraint violation bounds of $O(\sqrt{T})$ and $\widetilde{O}(\sqrt{T})$, respectively, for convex losses (Theorem 5), and improves the regret bound to $O(\log T)$

[2] By nearly optimal guarantees, we mean regret and violation bounds of order $\widetilde{O}(\sqrt{T})$.

Table 1: Comparison of COCO algorithms with optimal guarantees up to logarithmic factors. The columns represent the following. **Stochastic**: the stochastic constraint setting is considered; **Exp.**, **H.P.**: expected and high-probability guarantees are provided, respectively; **Adversarial**: the adversarial constraint setting is considered; **w/o Slater**: the algorithm does not require Slater's condition, where △ denotes that another regularity assumption, weaker than Slater's condition, is required; **S.C.**: improved guarantees can be achieved when losses are strongly convex; **Anytime**: the algorithm does not require T as an input parameter. †: Yu et al. (2017) considered only stochastic constraints, but the same algorithm also works for adversarial constraints (Neely and Yu, 2017).

Algorithm	Stochastic		Adversarial	w/o Slater	S.C.	Anytime
	Exp.	H.P.				
Yu et al. (2017)	✓	✓	✓†			
Wei et al. (2020)	✓			△		
Sinha and Vaze (2024)			✓	✓	✓	
Sarkar and Sinha (2026b)			✓	✓		✓
Ours	✓	✓	✓	✓	✓	✓

for strongly convex losses (Theorem 6). Here, hard constraint violation is a stronger notion of constraint violation that does not allow cancellation across rounds.

- Our algorithm is unified and anytime. The same algorithmic framework, based on the dual update with a regularizer, applies to both stochastic and adversarial constraint settings, as well as to both convex and strongly convex losses. Moreover, the parameter choices of our algorithm do not require prior knowledge of the horizon T, and hence the algorithm runs in an anytime manner without using the doubling trick.

The rest of the paper is organized as follows. Section 1.2 reviews additional related work. Section 2 presents the problem formulations and assumptions. Section 3 introduces our anytime primal-dual algorithm, and Section 4 explains the intuition for how the regularizer stabilizes the dual process. Section 5 presents the main theoretical results for stochastic and adversarial constraints under convex and strongly convex losses, together with an overview of the proofs. Sections 6 and 7 provide the analyses for expected and high-probability guarantees under stochastic constraints, respectively, and Section 8 extends the analysis to strongly convex losses. Section 9 concludes the paper, and the appendix contains the deferred proofs, the analysis for adversarial constraints, and auxiliary lemmas.

1.2 Related Work

Online Convex Optimization (OCO) provides a general framework for sequential decision-making with convex losses over a fixed feasible set. In the classical OCO setting, a decision maker selects an action in each round and then observes a convex loss function, with the goal of minimizing regret against a fixed comparator in hindsight. Since the foundational work of

Zinkevich (2003), OCO has been extensively studied in the online learning literature (Cesa-Bianchi et al., 1996; Cesa-Bianchi and Lugosi, 2006; Shalev-Shwartz, 2012; Hazan, 2016; Orabona, 2019).

In contrast to the classical OCO setting, COCO captures more complex feasible regions characterized by functional constraints and additionally requires the decision maker to control constraint violations over time. A line of work studies COCO with fixed or known constraint functions. A common formulation in this line is OCO with long-term constraints, where the constraints may be violated at individual rounds but their cumulative violation should be controlled over the entire horizon. This formulation was initiated by Mahdavi et al. (2012). Subsequent works developed adaptive algorithms (Jenatton et al., 2016), established guarantees for cumulative squared constraint violations (Yuan and Lamperski, 2018), and improved guarantees (Yu and Neely, 2020; Yi et al., 2021).

Another line of work studies COCO with stochastic constraints, where the constraint functions are sampled i.i.d. from an underlying distribution while the loss functions may vary adversarially. In this literature, Yu et al. (2017) established expected and high-probability regret and constraint violation bounds, but their analysis relies on Slater's condition. Later, Wei et al. (2020) relaxed this requirement by assuming the existence of bounded Lagrange multipliers. In contrast, our work requires none of these assumptions but still achieves nearly optimal guarantees for stochastic constraints.

Recently, COCO has also been extensively studied under adversarial constraints, where the loss and constraint functions may vary adversarially over time. The early work of Mannor et al. (2009) considered online learning with a comparator that satisfies a sample path constraint $\sum_{t=1}^{T} g_t(x) \leq 0$. They proved that for their setting, no online algorithm can achieve sublinear bounds on both regret and constraint violation at the same time. Subsequent works then studied adversarial constraints under restrictive comparators with round-wise feasibility requirements (Neely and Yu, 2017; Guo et al., 2022; Sinha and Vaze, 2024; Lekeufack and Jordan, 2024). This line of research has since been extended to broader settings (Wang et al., 2025a; Sinha and Vaze, 2026; Zhang, 2026; Supantha and Sinha, 2026). In particular, Garber and Kretzu (2024); Wang et al. (2025b); Sarkar et al. (2026) proposed projection-free algorithms, Yi et al. (2020, 2022); Pan et al. (2026) studied distributed settings, Vaze and Sinha (2026) proposed instance-dependent guarantees, Sarkar and Sinha (2026a) improved guarantees, and Sarkar and Sinha (2026b) proposed anytime algorithms. Although these results provide strong guarantees in the adversarial constraint setting, they do not directly apply to the stochastic constraint setting, where the comparator is required to be feasible only in expectation.

2 Preliminaries

This section presents the formulations of COCO studied in this paper and the required assumptions. Before introducing them, we first establish some basic notation used throughout the paper. Let $\mathbb{E}$ denote the expectation taken over randomness of the constraint functions, let $\mathbb{R}_+$ denote the set of nonnegative real numbers, let $\|\cdot\|$ denote the Euclidean norm, and let $[\cdot]_+ = \max\{0, \cdot\}$. Moreover, for a subdifferentiable function $f : \mathbb{R}^d \to \mathbb{R}$, let $\nabla f(\mathbf{x})$ denote a subgradient of f at $\mathbf{x}$. For a positive integer T, let $[T] = \{1, 2, \ldots, T\}$.

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a compact convex set for some positive integer d. Let $\{f_t\}_{t\geq 1}$ and $\{g_t\}_{t\geq 1}$ denote sequences of convex loss and constraint functions, respectively, that are subdifferentiable on $\mathcal{X}$. Additionally, we impose the following assumptions throughout the paper.

Assumption 1. There exists a constant $D > 0$ such that $\|\mathbf{x} - \mathbf{y}\| \leq D$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$.

Assumption 2. There exist constants $L, G > 0$ such that $\|\nabla f_t(\mathbf{x})\| \leq L$, $\|\nabla g_t(\mathbf{x})\| \leq L$, and $|g_t(\mathbf{x})| \leq G$ for all $\mathbf{x} \in \mathcal{X}$ and $t \geq 1$.

The first assumption ensures that $\mathcal{X}$ is bounded, while the second assumes that the subgradients and constraint functions are bounded. Both assumptions are common in the COCO literature (Yu et al., 2017).

We now formulate an online learning problem for COCO. In each round t, the decision maker selects an action $\mathbf{x}_t \in \mathcal{X}$, after which the loss function f_t and the constraint function g_t are revealed. We consider two settings that differ in how the constraint functions are given: the stochastic constraint setting (Yu et al., 2017) and the adversarial constraint setting (Sinha and Vaze, 2024).

- **Stochastic Constraint Setting:** In each round t, g_t is sampled i.i.d. from an unknown distribution $\mathcal{D}$, while f_t is chosen arbitrarily and independently of the samples of the constraint functions. In this case, we consider the following performance metrics, which are regret and constraint violation over T rounds:

$$\text{Regret}(T) = \sum_{t=1}^{T} (f_t(\mathbf{x}_t) - f_t(\mathbf{x}^\star)), \quad \text{Violation}(T) = \sum_{t=1}^{T} g_t(\mathbf{x}_t), \tag{1}$$

  where the comparator $\mathbf{x}^\star \in \mathcal{X}$ is independent of $\{g_t\}_{t=1}^T$ and is defined to be feasible in expectation, i.e.,

$$\mathbf{x}^\star \in \arg\min_{\mathbf{x}\in\mathcal{X}} \sum_{t=1}^{T} f_t(\mathbf{x}) \quad \text{s.t.} \quad \mathbb{E}[g_1(\mathbf{x})] = \cdots = \mathbb{E}[g_T(\mathbf{x})] \leq 0. \tag{2}$$

- **Adversarial Constraint Setting:** In each round t, g_t is chosen adversarially, just as f_t is. In this case, we consider the regret measure defined in the same way as in (1), while the notion of hard constraint violation is considered, which is defined as

$$\text{Violation}_+(T) = \sum_{t=1}^{T} [g_t(\mathbf{x}_t)]_+.$$

  Here, under the assumption that $\{\mathbf{x} \in \mathcal{X} : g_t(\mathbf{x}) \leq 0, \ \forall t \in [T]\}$ is nonempty (Sinha and Vaze, 2024), $\mathbf{x}^\star$ is defined as round-wise feasible, i.e.,

$$\mathbf{x}^\star \in \arg\min_{\mathbf{x}\in\mathcal{X}} \sum_{t=1}^{T} f_t(\mathbf{x}) \quad \text{s.t.} \quad g_t(\mathbf{x}) \leq 0, \ \forall t \in [T]. \tag{3}$$

Algorithm 1 COCO without Slater's condition

Input: Lipschitz constant L; boundedness constant for constraints G; diameter D; confidence parameter $\delta \in (0,1)$; strong convexity constant μ; auxiliary function $\Psi : \mathbb{R}_+ \to \mathbb{R}$; Lyapunov parameters $\{\gamma_t\}_{t\geq 1}$.

Initialize: Set $\mathbf{x}_1 \in \mathcal{X}$, $Q_1 = 0$, and $\Phi_t(x) = e^{\gamma_t x} - 1$ for all $t \geq 1$.

for $t = 1, 2, \ldots$ **do**

Take $\mathbf{x}_t$ and observe f_t, g_t

Update the primal step size η_t

Update $\mathbf{x}_{t+1} = \mathrm{proj}_{\mathcal{X}}\left(\mathbf{x}_t - \eta_t(\nabla f_t(\mathbf{x}_t) + \Phi'_t(Q_t)\nabla g_t(\mathbf{x}_t))\right)$

Update R_t:

$$R_t = 12\gamma_t G^2 + \frac{\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)}{\Phi'_t(Q_t)}$$

Update $Q_{t+1} = Q_t + g_t(\mathbf{x}_t) - R_t$

end for

Again, it is worth comparing the definitions of $\mathbf{x}^\star$ in the two settings: feasible in expectation (2) versus round-wise feasible (3). In particular, for the adversarial constraint setting, $\mathbf{x}^\star$ is defined to be round-wise feasible, since no online algorithm can achieve sublinear bounds on both regret and constraint violation at the same time if $\mathbf{x}^\star$ is instead defined as $\arg\min_{\mathbf{x}\in\mathcal{X}} \sum_{t=1}^{T} f_t(\mathbf{x})$ s.t. $\sum_{t=1}^{T} g_t(\mathbf{x}) \leq 0$ (Mannor et al., 2009). Moreover, under the restrictive definition of $\mathbf{x}^\star$ given in (3), we may obtain guarantees for the hard constraint violation. On the other hand, under the stochastic constraint setting, $\mathbf{x}^\star$ is only needed to satisfy the constraint in expectation rather than for all rounds. Therefore, the difference between two choices of $\mathbf{x}^\star$ suggests that algorithms for the adversarial constraint setting do not directly translate to the stochastic constraint setting, requiring additional techniques.

3 Unified Algorithm for COCO

In this section, we present Algorithm 1, an anytime algorithm for solving COCO. We emphasize that it is a unified algorithm that does not rely on Slater's condition and achieves nearly optimal guarantees in both stochastic and adversarial constraint settings under both convex and strongly convex losses. The key principle is to incorporate an adaptive regularizer into the dual update that stabilizes the dual process without relying on the negative drift induced by Slater's condition. A more detailed discussion of the regularizer is provided in Section 4.

Our algorithm proceeds as follows; the choices of Ψ, γ_t, and η_t will be given in the subsequent paragraph. Initially, the decision maker selects an arbitrary point $\mathbf{x}_1 \in \mathcal{X}$ and sets the initial dual variable $Q_1 = 0$. In each round $t \geq 1$, the decision maker first plays $\mathbf{x}_t$ and then observes the loss function f_t and the constraint function g_t. Using the current dual variable Q_t, inspired by the works of Sinha and Vaze (2024); Sarkar and Sinha (2026b), the algorithm constructs an exponential Lyapunov function given by $\Phi_t(x) = e^{\gamma_t x} - 1$ and performs a primal update based on the surrogate subgradient $\nabla f_t(\mathbf{x}_t) + \Phi'_t(Q_t)\nabla g_t(\mathbf{x}_t)$. The exponential Lyapunov function is particularly useful compared to quadratic Lyapunov functions because its derivative is strictly positive for all Q_t when $\gamma_t > 0$, implying that the surrogate function $f_t + \Phi_t'(Q_t)g_t$ always remains convex. The dual variable is then updated according to the observed constraint violation together with the regularizer term R_t, which adaptively controls its growth.

The key parameters of Algorithm 1 are Ψ, γ_t, and η_t. Indeed, these parameters do not require knowledge of T, hence our algorithm is anytime. The choices of Ψ, γ_t, and η_t are given as follows, depending on the setting under consideration.

- The parameter γ_t defines the exponential Lyapunov function Φ_t, while Ψ is an auxiliary function used to define R_t. Under the stochastic constraint setting, both Ψ and γ_t are chosen according to the type of guarantee desired, namely, expected or high-probability guarantees. The precise choices of Ψ and γ_t are given as follows: if the expected guarantees are desired, then we take

$$\Psi(x) = 4DL\sqrt{x}, \qquad \gamma_t = \min\left\{\frac{1}{12G\sqrt{t}}, \frac{1}{24DL}, 1\right\}. \tag{4}$$

  To obtain high-probability guarantees with confidence $\delta \in (0, 1)$, we instead take

$$\begin{aligned} \Psi(x) &= 4DL\sqrt{x} + 4G\sqrt{(1+x)\log\frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}}, \\ \gamma_t &= \min\left\{\frac{1}{12G\sqrt{t}}, \frac{1}{24\Big(DL + 8G\sqrt{\log(12t^2/\delta)}\Big)}, 1\right\}. \end{aligned} \tag{5}$$

  For the adversarial constraint setting, we use the same choices of Ψ and γ_t as in the expected bound setting with stochastic constraints. Moreover, with a minor modification, the algorithm can also provide guarantees for hard constraint violation. The modified algorithm is given in Algorithm 2, and the corresponding results are presented in Section 5.3.

- The primal step size η_t is used in the projected subgradient descent update and is chosen according to the properties of the loss functions. Specifically, when the loss functions are convex, we employ an AdaGrad-type step size for η_t (Duchi et al., 2011). In contrast, to deduce improved performance when the loss functions are strongly convex, we use a step size based on that of Bartlett et al. (2007), with appropriate modifications to accommodate our setting. The precise choice of η_t is summarized as follows: if losses are convex, then we take

$$\eta_t = \frac{\sqrt{2}D}{2\sqrt{1 + \sum_{\tau=1}^{t} \|\nabla f_\tau(\mathbf{x}_\tau) + \Phi_\tau'(Q_\tau)\nabla g_\tau(\mathbf{x}_\tau)\|^2}}. \tag{6}$$

  If losses are μ-strongly convex, then we take

$$\eta_t = \frac{1}{\mu t + (2L/D)\sqrt{\sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)^2}}. \tag{7}$$

4 The Role of the Regularizer

This section explains the role of the regularizer R_t. In the stochastic constraint setting, the main challenge arises from the less restrictive definition of comparator $\mathbf{x}^\star$, that is, a solution satisfying the expected constraint $\mathbb{E}[g_t(\mathbf{x}^\star)] \le 0$. Specifically, although the algorithm proposed by Sinha and Vaze (2024) attained nearly optimal guarantees in the adversarial constraint setting, its analysis heavily relies on the fact that $g_t(\mathbf{x}^\star) \le 0$ for all $t \ge 1$, which no longer holds in our setting. To address this issue, our technique is to incorporate an adaptive regularizer into the dual update.

4.1 Key Mechanism

Before explaining how the regularizer works, we first introduce the main direction of the analysis. Inspired by Sinha and Vaze (2024), we aim to show the following key relation, which simultaneously guarantees small regret and stability of the dual process. Letting $\lesssim$ denote an informal notation used only to convey intuition,

$$\mathrm{Regret}(T) + \Phi_{T+1}(Q_{T+1}) \lesssim O(\sqrt{T}). \tag{8}$$

To derive this relation in the stochastic constraint setting, we then explain why the regularizer is essential. For this purpose, we first observe that the regularizer enables us to control the Lyapunov drift more carefully, which is defined as $\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t)$. Intuitively, since this quantity represents the change in the dual variable, it is desirable for it to remain small in order to stabilize the dual process. Roughly, the Lyapunov drift satisfies the following relation, suggesting that R_t helps control the Lyapunov drift and thereby stabilizes the dual process.

$$\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t) \lesssim \Phi'_t(Q_t)\big(g_t(\mathbf{x}_t) - R_t\big). \tag{9}$$

Combining this drift analysis with the regret analysis for the surrogate loss $f_t + \Phi'_t(Q_t)g_t$ gives us

$$\begin{aligned} &\mathrm{Regret}(T) + \Phi_{T+1}(Q_{T+1}) \\ &\lesssim O\left(\sqrt{T}\right) + \underbrace{\sqrt{\sum_{t=1}^{T} \Phi'_t(Q_t)^2}}_{\text{(I)}} - \underbrace{\sum_{t=1}^{T} \Phi'_t(Q_t) R_t}_{\text{(II)}} + \underbrace{\sum_{t=1}^{T} \Phi'_t(Q_t) g_t(\mathbf{x}^\star)}_{\text{(III)}}. \end{aligned} \tag{10}$$

This relation clarifies how R_t works. In particular, to deduce (8) from the above relation, the most problematic term is (I), whereas (III)—Lyapunov-weighted sum of comparator constraints—is nonpositive in expectation from the feasibility of $\mathbf{x}^\star$. We emphasize that, by adopting a carefully designed regularizer, (II) can offset this problematic term (I), resulting in (8). This observation motivates us to choose R_t as

$$R_t \approx \frac{\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} - \sqrt{\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2}}{\Phi'_t(Q_t)},$$

which justifies the choice of R_t. Without the regularizer, it is nontrivial to bound (I), which highlights that our regularizer is essential for deriving small regret and stability of the dual process.

So far, we have demonstrated the mechanism of R_t in the dual update. Although adopting R_t is useful for achieving small regret and stability of the dual process, there is another caveat in the choice of R_t. That is, when R_t is too large, the constraint violation cannot be bounded, based on the observation that $\text{Violation}(T) = Q_{T+1} + \sum_{t=1}^{T} R_t$. Hence, there is a trade-off in the choice of R_t in the sense that it is helpful to stabilize the dual process but, at the same time, it could harm the violation of the constraint. Nevertheless, under our choice of R_t, we prove that its cumulative sums are sublinear, e.g., Lemma 11. Consequently, our choice of R_t yields the desired nearly optimal guarantees without Slater's condition.

4.2 Comparison with the Existing Dual Updates

In this subsection, we compare our algorithm with the algorithms of Sinha and Vaze (2024) and Yu et al. (2017). We first discuss the relation to Sinha and Vaze (2024), which studied the adversarial constraint setting without Slater's condition. In particular, we focus on addressing the following question: why is R_t unnecessary in the adversarial setting, whose dual update is $Q_t = Q_{t-1} + [g_t(\mathbf{x}_t)]_+$? To see this, we provide the analogue of (10), which can be written as

$$\text{Regret}(T) + \Phi(Q_T) \lesssim O\left(\sqrt{T}\right) + \underbrace{\sqrt{\sum_{t=1}^{T} \Phi'(Q_t)^2}}_{(\text{I}')} + \underbrace{\sum_{t=1}^{T} \Phi'(Q_t)[g_t(\mathbf{x}^\star)]_+}_{(\text{III}')} . \tag{11}$$

Since their dual update ensures that $\{Q_t\}_{t\geq 1}$ is an increasing sequence, choosing $\Phi(x) = e^{\gamma x} - 1$ with $\gamma = 1/(2\sqrt{T})$ yields $(\text{I}') \leq \Phi(Q_T)/2$. Hence, (I') can be absorbed into $\Phi(Q_T)$ on the left-hand side, without relying on a regularizer. Moreover, (III') is 0 under their choice of $\mathbf{x}^\star$, since $g_t(\mathbf{x}^\star) \leq 0$ for all t. These observations explain why the algorithm of Sinha and Vaze (2024) works without a regularizer. However, this approach no longer holds in our setting. The main reason is that since we only have $\mathbb{E}[g_t(\mathbf{x}^\star)] \leq 0$ in the stochastic constraint setting, (III') does not coincide with 0, even after taking expectations. In summary, in the adversarial constraint setting, (8) is attained without a regularizer, and this is possible due to the restrictive definition of $\mathbf{x}^\star$. This highlights both the challenge in the stochastic constraint setting and the importance of adopting R_t.

We conclude the section by comparing our approach with Yu et al. (2017), which studies the same setting as ours but requires Slater's condition. This comparison explains why our approach does not require Slater's condition. To see this, under the quadratic Lyapunov function $\Phi(x) = x^2/2$ and the dual update $Q_{t+1} = [Q_t + g_t(\mathbf{x}_t) + \nabla g_t(\mathbf{x}_t)^\top(\mathbf{x}_{t+1} - \mathbf{x}_t)]_+$, their analysis shows that the conditional expected Lyapunov drift can be bounded as follows (see the proof of Lemma 7 in Yu et al. (2017)):

$$\mathbb{E}[\Phi(Q_{t+t_0}) - \Phi(Q_t)|\mathcal{F}_{t-1}] \lesssim -\varepsilon t_0 \Phi'(Q_t) + C \tag{12}$$

where ε is a Slater constant, t_0 is a window parameter, $\mathcal{F}_{t-1}$ is the σ-algebra generated by the history up to round $t-1$, and C collects the remaining positive terms. This shows that the stability of their dual process stems from $-\varepsilon t_0 \Phi'(Q_t)$, as it is the only negative term on the right-hand side. Therefore, their analysis necessitates Slater's condition. In contrast, our analysis relies on the regularizer as in (9). This is the key reason why our algorithm is free from Slater's condition.

5 Main Results

In this section, we present the theoretical guarantees of our algorithm. In particular, we establish guarantees for both convex and strongly convex loss functions under stochastic and adversarial constraints. For the stochastic constraint setting, we further provide both expected and high-probability guarantees. An overview of the analysis is presented in Section 5.4.

5.1 Stochastic Constraints

We first consider the stochastic constraint setting, where the loss functions may vary adversarially over time while the constraint functions are stochastic. Throughout this subsection, both the loss and constraint functions are assumed to be convex. To attain expected guarantees in this setting, recall that we choose the auxiliary function Ψ, the Lyapunov parameter γ_t as in (4), and the primal step size η_t as in (6). We are now ready to present our first theoretical guarantee, where the corresponding analysis is provided in Section 6.

Theorem 1. *Consider convex losses and stochastic constraints. For any $T \geq 1$, suppose that we run Algorithm 1 for T rounds with Ψ, γ_t given by (4) and η_t given by (6). Then, we have*

$$\mathbb{E}[\mathrm{Regret}(T)] = O\left(\sqrt{T}\right), \quad \mathbb{E}[\mathrm{Violation}(T)] = O\left(\sqrt{T}\log T\right),$$

where the expectation is taken with respect to the randomness of $\{g_t\}_{t\geq 1}$.

We note that this theorem demonstrates that our algorithm achieves nearly optimal expected guarantees in T, without relying on Slater's condition.

While Theorem 1 attains guarantees in expectation, it is often desirable to obtain realized guarantees that hold with high confidence. Thus, given a confidence parameter $\delta \in (0,1)$, we establish high-probability guarantees under the same setting as in Theorem 1. To this end, we modify Ψ, γ_t as in (5), while retaining the same primal step size η_t as in (6). Under these modified algorithmic parameters, the algorithm enjoys the following high-probability guarantees. The analysis of the following theorem is presented in Section 7.

Theorem 2. *Consider convex losses and stochastic constraints. For any $T \geq 1$ and $\delta \in (0,1)$, suppose that we run Algorithm 1 for T rounds with Ψ, γ_t given by (5) and η_t given by (6). Then, with probability at least $1-\delta$, we have*

$$\mathrm{Regret}(T) = O\left(\sqrt{T} + \sqrt{\log(1/\delta)}\right), \quad \mathrm{Violation}(T) = O\left(\sqrt{T}\log(T/\delta) + \log^{3/2}(T/\delta)\right).$$

5.2 Extension to Strongly Convex Losses

We next extend the preceding results to the setting with μ-strongly convex losses, while the remaining assumptions are identical to those in Section 5.1. To obtain improved guarantees, it is sufficient to modify the primal step size η_t as in (7), which more properly exploits strong convexity (Lemma 17). We then obtain the following expected and high-probability guarantees.

Theorem 3. *Consider μ-strongly convex losses for $\mu > 0$ and stochastic constraints. For any $T \geq 1$, suppose that we run Algorithm 1 for T rounds with Ψ, γ_t given by (4) and η_t given by (7). Then, we have*

$$\mathbb{E}[\mathrm{Regret}(T)] = O\left(\log T\right), \quad \mathbb{E}[\mathrm{Violation}(T)] = O\left(\sqrt{T}\log T\right),$$

where the expectation is taken with respect to the randomness of $\{g_t\}_{t\geq 1}$.

Theorem 4. *Consider μ-strongly convex losses for $\mu > 0$ and stochastic constraints. For all $T \geq 1$ and $\delta \in (0,1)$, suppose that we run Algorithm 1 for T rounds with Ψ, γ_t given by (5) and η_t given by (7). Then, with probability at least $1-\delta$, we have*

$$\mathrm{Regret}(T) = O\left(\log T + \sqrt{\log(1/\delta)}\right), \quad \mathrm{Violation}(T) = O\left(\sqrt{T}\log(T/\delta) + \log^{3/2}(T/\delta)\right).$$

Both theorems demonstrate that our algorithm can achieve improved expected and high-probability regret bounds that scale as $\log T$ when the losses are strongly convex. The analyses of Theorems 3 and 4 are presented in Section 8.

5.3 Adversarial Constraints

Finally, we consider the adversarial constraint setting. Although Sinha and Vaze (2024) already established nearly optimal guarantees in this setting, the following results demonstrate that our framework is unified in the sense that it achieves nearly optimal guarantees under both stochastic and adversarial constraints.[3]

Although we may directly apply Algorithm 1 to the adversarial constraint setting, the resulting guarantees are on cumulative constraint violation, given by $\mathrm{Violation}(T) = \sum_{t=1}^{T} g_t(\mathbf{x}_t)$. Since this quantity allows cancellation between positive and negative violations across time, it is weaker than the notion of hard constraint violation, given by $\mathrm{Violation}_+(T) = \sum_{t=1}^{T}[g_t(\mathbf{x}_t)]_+$, which is often the performance metric adopted by adversarial-constraint algorithms. To obtain guarantees with respect to $\mathrm{Violation}_+(T)$, we introduce several modifications to Algorithm 1.

The only difference is that Algorithm 2 uses $\tilde{g}_t$ in place of g_t, where $\tilde{g}_t : \mathbf{x} \mapsto [g_t(\mathbf{x})]_+$, and $\nabla\tilde{g}_t(\mathbf{x}_t)$ denotes a subgradient of $\tilde{g}_t$. Note that this modification yields hard constraint violation guarantees, since $\tilde{g}_t(\mathbf{x}_t)$ is accumulated in the dual variable update and $\sum_{t=1}^{T}\tilde{g}_t(\mathbf{x}_t)$ coincides with $\mathrm{Violation}_+(T)$. Hence, with Algorithm 2, we can obtain guarantees for $\mathrm{Violation}_+(T)$ by applying the same analysis strategy used for the stochastic constraint setting. This is possible because the comparator $\mathbf{x}^\star$ in the adversarial constraint setting is feasible in every round, i.e., $g_t(\mathbf{x}^\star) \leq 0$ for all $t \geq 1$, which is stronger than feasibility

[3] Recently, this line of work has been extended to various settings (Sinha and Vaze, 2026; Sarkar et al., 2026; Sarkar and Sinha, 2026a; Vaze and Sinha, 2026; Supantha and Sinha, 2026). In contrast, our focus is to demonstrate that the proposed algorithm applies to both stochastic and adversarial constraint settings. We leave extensions of our algorithm to these broader settings for future work.

Algorithm 2 COCO for Hard Constraint Violation without Slater's condition

Input: Lipschitz constant L; boundedness constant for constraints G; diameter D; confidence parameter $\delta \in (0,1)$; strong convexity constant μ; auxiliary function $\Psi : \mathbb{R}_+ \to \mathbb{R}$; Lyapunov parameters $\{\gamma_t\}_{t\geq 1}$.

Initialize: Set $\mathbf{x}_1 \in \mathcal{X}$, $Q_1 = 0$, and $\Phi_t(x) = e^{\gamma_t x} - 1$ for all $t \geq 1$.

for $t = 1, 2, \ldots$ **do**

Take $\mathbf{x}_t$ and observe f_t, g_t

Set $\tilde{g}_t \leftarrow [g_t]_+$ and $\nabla \tilde{g}_t(\mathbf{x}_t) = \nabla g_t(\mathbf{x}_t)$ if $g_t(\mathbf{x}_t) > 0$, and $\mathbf{0}$ otherwise

Update the primal step size η_t

Update $\mathbf{x}_{t+1} = \mathrm{proj}_{\mathcal{X}}\left(\mathbf{x}_t - \eta_t(\nabla f_t(\mathbf{x}_t) + \Phi_t'(Q_t)\nabla \tilde{g}_t(\mathbf{x}_t))\right)$

Update R_t:

$$R_t = 12\gamma_t G^2 + \frac{\Psi\left(\sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi_\tau'(Q_\tau)^2\right)}{\Phi_t'(Q_t)}$$

Update $Q_{t+1} = Q_t + \tilde{g}_t(\mathbf{x}_t) - R_t$

end for

in expectation used in the stochastic constraint setting. Finally, we obtain the following results, whose analyses are deferred to Appendix B, as they follow from straightforward modifications of the previous analysis.

Theorem 5. *Consider convex losses and adversarial constraints. For all $T \geq 1$, suppose that we run Algorithm 2 for T rounds with Ψ, γ_t given by (4) and η_t given by (6). Then, we have*

$$\mathrm{Regret}(T) = O\left(\sqrt{T}\right), \quad \mathrm{Violation}_+(T) = O\left(\sqrt{T}\log T\right).$$

Theorem 6. *Consider μ-strongly convex losses for $\mu > 0$ and adversarial constraints. For all $T \geq 1$, suppose that we run Algorithm 2 for T rounds with Ψ, γ_t given by (4) and η_t given by (7). Then, we have*

$$\mathrm{Regret}(T) = O\left(\log T\right), \quad \mathrm{Violation}_+(T) = O\left(\sqrt{T}\log T\right).$$

5.4 Overview of Proofs

The analyses of all theorems follow the same template, while additional techniques are required for handling high-probability guarantees and strongly convex losses. First, we establish a drift bound for the time-varying exponential Lyapunov function. Second, we combine this drift bound with a surrogate regret guarantee with respect to $f_t + \Phi_t'(Q_t)g_t$. Third, the regularizer cancels the accumulated term involving $\sum_{t=1}^{T} \Phi_t'(Q_t)^2$. The remaining work is to control the Lyapunov-weighted comparator constraint term, which is nonpositive in expectation and controlled by a supermartingale concentration inequality with high probability, and then to bound the cumulative sum of regularizers $\sum_{t=1}^{T} R_t$.

6 Analysis under Stochastic Constraints: Expected Bounds

In this section, we present the analysis of Theorem 1, which establishes expected guarantees under convex losses and stochastic constraints. Throughout, $\mathbb{E}$ denotes the expectation taken with respect to the randomness of $\{g_t\}_{t\geq 1}$, and $\mathcal{F}_t$ denotes the σ-algebra generated by $\{f_\tau\}_{\tau\geq 1}$ and $\{g_\tau\}_{\tau=1}^t$.

Lemma 7. Suppose that we run Algorithm 1 with γ_t, Ψ as given by (4). Then, for all $t \geq 1$, we have

$$\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t) \leq \Phi_t'(Q_t)\left(g_t(\mathbf{x}_t) - \frac{R_t}{2}\right) + \frac{\gamma_t - \gamma_{t+1}}{e\gamma_{t+1}}.$$

Proof. Note that the Lyapunov drift can be decomposed as

$$\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t) = \underbrace{\Phi_t(Q_{t+1}) - \Phi_t(Q_t)}_{\text{(I)}} + \underbrace{\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_{t+1})}_{\text{(II)}}. \tag{13}$$

Now, we first bound (I). Recall that $\Phi_t(x) = e^{\gamma_t x} - 1$ and $\Phi_t'(x) = \gamma_t e^{\gamma_t x}$. It follows that

$$\begin{aligned}\Phi_t(Q_{t+1}) - \Phi_t(Q_t) &= e^{\gamma_t Q_{t+1}} - e^{\gamma_t Q_t} \\ &= e^{\gamma_t Q_t}(e^{\gamma_t(Q_{t+1}-Q_t)} - 1) \\ &= e^{\gamma_t Q_t}(e^{\gamma_t(g_t(\mathbf{x}_t)-R_t)} - 1) \\ &= \frac{1}{\gamma_t}\Phi_t'(Q_t)(e^{\gamma_t(g_t(\mathbf{x}_t)-R_t)} - 1).\end{aligned}$$

Note that $e^x \leq 1 + x + x^2$ for all $x \leq 1$, $\gamma_t(g_t(\mathbf{x}_t) - R_t) \leq \gamma_t G \leq 1$, and $\Phi_t'(Q_t)/\gamma_t > 0$. Then it follows that

$$\begin{aligned}&\Phi_t(Q_{t+1}) - \Phi_t(Q_t) \\ &\leq \frac{1}{\gamma_t}\Phi_t'(Q_t)\left(\gamma_t(g_t(\mathbf{x}_t) - R_t) + \gamma_t^2(g_t(\mathbf{x}_t) - R_t)^2\right) \qquad (14) \\ &\leq \Phi_t'(Q_t)\left(g_t(\mathbf{x}_t) - R_t + 3\gamma_t G^2 + 432\gamma_t^3 G^4 + 3\gamma_t\left(\frac{\Psi\left(\sum_{\tau=1}^t \Phi_\tau'(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi_\tau'(Q_\tau)^2\right)}{\Phi_t'(Q_t)}\right)^2\right)\end{aligned}$$

where the second inequality is due to the fact that $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ for any $x, y, z \in \mathbb{R}$. Since $\gamma_t \leq 1/(12G)$, we have

$$3\gamma_t G^2 + 432\gamma_t^3 G^4 \leq 3\gamma_t G^2 + 432\gamma_t G^4 \cdot \frac{1}{144G^2} = 6\gamma_t G^2. \tag{15}$$

Moreover, we have

$$
\begin{aligned}
&3\gamma_t \frac{\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)}{\Phi'_t(Q_t)} \\
&= 12\gamma_t DL \frac{\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} - \sqrt{\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2}}{\Phi'_t(Q_t)} \\
&= \frac{12\gamma_t DL \Phi'_t(Q_t)}{\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + \sqrt{\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2}} \\
&\le 12\gamma_t DL \\
&\le \frac{1}{2}
\end{aligned}
\tag{16}
$$

where the first inequality follows from $\Phi'_t(Q_t) \le \sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + \sqrt{\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2}$, and the second inequality is due to $\gamma_t \le 1/(24DL)$. Note that Ψ is an increasing function, implying that $\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)$ is nonnegative. Then, by applying (15) and (16) to (14), (I) is bounded as

$$
\begin{aligned}
\text{(I)} &= \Phi_t(Q_{t+1}) - \Phi_t(Q_t) \\
&\le \Phi'_t(Q_t)\left(g_t(\mathbf{x}_t) - R_t + 6\gamma_t G^2 + \frac{\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)}{2\Phi'_t(Q_t)}\right) \\
&= \Phi'_t(Q_t)\left(g_t(\mathbf{x}_t) - \frac{1}{2}R_t\right).
\end{aligned}
$$

Next, we bound (II). Since $\gamma_{t+1} \le \gamma_t$, by Lemma 29, we have

$$
\text{(II)} = \Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_{t+1}) \le \frac{\gamma_t - \gamma_{t+1}}{e\gamma_{t+1}}.
$$

By applying bounds on (I) and (II) to (13), we conclude the proof. □

From the right-hand side of the inequality in Lemma 7, we observe that the cost of employing time-varying Lyapunov functions is $(\gamma_t - \gamma_{t+1})/(e\gamma_{t+1})$. Indeed, the following lemma demonstrates that this cost is not significant by showing that its summation over $\tau = 1, \ldots, t$ grows only logarithmically with t.

Lemma 8. Consider γ_t given by (4). Then, we have

$$
\sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{\gamma_{\tau+1}} \le \frac{1 + \log t}{2}.
$$

Proof. Recall that $1/\gamma_\tau = \max\{12G\sqrt{\tau}, 24DL, 1\}$. Then it follows that

$$
\frac{\gamma_\tau - \gamma_{\tau+1}}{\gamma_{\tau+1}} = \frac{\frac{1}{\gamma_{\tau+1}} - \frac{1}{\gamma_\tau}}{\frac{1}{\gamma_\tau}} \le \frac{\sqrt{\tau+1} - \sqrt{\tau}}{\sqrt{\tau}} = \frac{1}{(\sqrt{\tau+1} + \sqrt{\tau})\sqrt{\tau}} \le \frac{1}{2\tau}
$$

where the first inequality follows from $\max\{x, y, z\} - \max\{x', y, z\} \le x - x'$ for any $x > x'$ and $1/\gamma_\tau \ge 12G\sqrt{\tau}$. Thus, we have

$$\sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{\gamma_{\tau+1}} \le \sum_{\tau=1}^{t} \frac{1}{2\tau} \le \frac{1 + \log t}{2},$$

as required. $\square$

Next, we introduce Lemma 9, which provides an AdaGrad-type guarantee for the surrogate loss $f_t + \Phi_t'(Q_t)g_t$.

Lemma 9. Consider the setting of Theorem 1. Suppose that we run Algorithm 1 with η_t given by (6). Then, for all $t \ge 1$, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau) g_\tau(\mathbf{x}_\tau) \\
&\le \sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau) g_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + 2DL\sqrt{\sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)^2} + D.
\end{aligned}$$

Proof. For simplicity, let $\hat{f}_\tau = f_\tau + \Phi_\tau'(Q_\tau)g_\tau$ for each $\tau \in [t]$. Note that since $\Phi_\tau'(Q_\tau) > 0$, $\hat{f}_\tau$ is convex. Since projection onto $\mathcal{X}$ is a contraction mapping, we have for any $\mathbf{x}^\star \in \mathcal{X}$,

$$\begin{aligned}
\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2 &\le \|\mathbf{x}_\tau - \eta_\tau \nabla \hat{f}_\tau(\mathbf{x}_\tau) - \mathbf{x}^\star\|^2 \\
&\le \|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 + \eta_\tau^2 \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2 - 2\eta_\tau \nabla \hat{f}_\tau(\mathbf{x}_\tau)^\top (\mathbf{x}_\tau - \mathbf{x}^\star).
\end{aligned}$$

By convexity, we have $\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star) \le \nabla \hat{f}_\tau(\mathbf{x}_\tau)^\top (\mathbf{x}_\tau - \mathbf{x}^\star)$. It follows that

$$\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star) \le \frac{\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2}{2\eta_\tau} + \frac{\eta_\tau \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2}{2}.$$

Summing over $\tau = 1, \ldots, t$ yields

$$\sum_{\tau=1}^{t} (\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star)) \le \sum_{\tau=1}^{t} \frac{\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2}{2\eta_\tau} + \sum_{\tau=1}^{t} \frac{\eta_\tau \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2}{2}.$$

Note that

$$\begin{aligned}
\sum_{\tau=1}^{t} \frac{\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2}{2\eta_\tau} &\le \sum_{\tau=1}^{t-1} \frac{\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2}{2} \left(\frac{1}{\eta_{\tau+1}} - \frac{1}{\eta_\tau} \right) + \frac{D^2}{2\eta_1} \\
&\le \frac{D^2}{2\eta_t} \\
&= \frac{D}{\sqrt{2}} \sqrt{1 + \sum_{\tau=1}^{t} \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2} \\
&\le \frac{D}{\sqrt{2}} \sqrt{\sum_{\tau=1}^{t} \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2} + \frac{D}{\sqrt{2}}
\end{aligned}$$

where the second inequality follows from $1/\eta_{\tau+1} - 1/\eta_\tau \geq 0$ and $\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2 \leq D^2$, and the last inequality follows from $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for any $a, b \geq 0$. Moreover,

$$\begin{aligned}
\sum_{\tau=1}^{t} \frac{\eta_\tau \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2}{2} &= \frac{D}{2\sqrt{2}} \sum_{\tau=1}^{t} \frac{\|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2}{\sqrt{1 + \sum_{s=1}^{\tau} \|\nabla \hat{f}_s(\mathbf{x}_s)\|^2}} \\
&\leq \frac{D}{2\sqrt{2}} \cdot 2 \left(\sqrt{1 + \sum_{\tau=1}^{t} \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2} - 1 \right) \\
&\leq \frac{D}{\sqrt{2}} \sqrt{\sum_{\tau=1}^{t} \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2}
\end{aligned}$$

where the second inequality follows from $\sum_{\tau=1}^{t} y_\tau / \sqrt{1 + \sum_{s=1}^{\tau} y_s} \leq 2(\sqrt{1 + \sum_{\tau=1}^{t} y_\tau} - 1)$ for any nonnegative sequence $\{y_\tau\}_{\tau \geq 1}$, and the last inequality follows from $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for any $a, b \geq 0$. Since $1/\sqrt{2} \leq 1$, it follows that

$$\sum_{\tau=1}^{t} (\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star)) \leq \sqrt{2} D \sqrt{\sum_{\tau=1}^{t} \|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2} + D.$$

Recall that $\hat{f}_\tau = f_\tau + \Phi'_\tau(Q_\tau) g_\tau$ and $\|\nabla f_\tau(\mathbf{x}_\tau)\|^2, \|\nabla g_\tau(\mathbf{x}_\tau)\|^2 \leq L^2$. It follows that

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)(g_\tau(\mathbf{x}_\tau) - g_\tau(\mathbf{x}^\star)) \\
&\leq 2DL\sqrt{t} + 2DL \sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + D.
\end{aligned}$$

By rearranging the above inequality, we conclude the proof. □

We now combine Lemmas 7 and 9. This yields the following lemma, which establishes both a small regret bound and the stability of the dual process in expectation.

Lemma 10. Consider the setting of Theorem 1. Suppose that we run Algorithm 1 with γ_t, Ψ given by (4) and η_t given by (6). Then, for all $t \geq 1$, we have

$$\mathbb{E}\left[\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star))\right] + \mathbb{E}\left[\Phi_{t+1}(Q_{t+1})\right] \leq 2DL\sqrt{t} + \log t + 1 + D.$$

Proof. By Lemma 9, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}_\tau) \\
&\leq \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + 2DL \sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + D.
\end{aligned}$$

By summing the inequality in Lemma 7 over $\tau = 1, \ldots, t$, since $\Phi_1(Q_1) = 0$, we have

$$\Phi_{t+1}(Q_{t+1}) - \Phi_1(Q_1) \leq \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) \left(g_\tau(\mathbf{x}_\tau) - \frac{R_\tau}{2} \right) + \sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}}.$$

Since $\Phi_1(Q_1) = 0$, it follows that

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \\
&\leq \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + 2DL\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + D + \sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} - \frac{1}{2}\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) R_\tau.
\end{aligned}$$

Note that

$$\begin{aligned}
\frac{1}{2}\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) R_\tau &\geq \frac{1}{2}\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) \cdot \frac{\Psi(\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2) - \Psi(\sum_{s=1}^{\tau-1} \Phi'_s(Q_s)^2)}{\Phi'_\tau(Q_\tau)} \\
&= \frac{1}{2}\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2 \right) - \frac{1}{2}\Psi(0) \\
&= 2DL\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2}.
\end{aligned}$$

Thus, we deduce that

$$\begin{aligned}
\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) &\leq \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + \sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} + D \\
&\leq \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + \log t + 1 + D.
\end{aligned}$$

where the second inequality follows from Lemma 8 and $1/(2e) \leq 1$. Now, taking expectations on both sides yields

$$\begin{aligned}
&\mathbb{E}\left[\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) \right] + \mathbb{E}\left[\Phi_{t+1}(Q_{t+1}) \right] \\
&\leq \sum_{\tau=1}^{t} \mathbb{E}\left[\Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) \right] + 2DL\sqrt{t} + \log t + 1 + D \\
&= \sum_{\tau=1}^{t} \mathbb{E}\left[\Phi'_\tau(Q_\tau) \mathbb{E}[g_\tau(\mathbf{x}^\star) | \mathcal{F}_{\tau-1}] \right] + 2DL\sqrt{t} + \log t + 1 + D \\
&\leq 2DL\sqrt{t} + \log t + 1 + D
\end{aligned}$$

where the equality follows from the tower rule and $\Phi'_\tau(Q_\tau)$ is $\mathcal{F}_{\tau-1}$-measurable, and the second inequality follows from the feasibility assumption that $\mathbb{E}[g_\tau(\mathbf{x}^\star)] \leq 0$ for fixed $\mathbf{x}^\star$ and that $\{g_\tau\}_{\tau \geq 1}$ are i.i.d. samples. This concludes the proof. $\square$

Next, we introduce Lemma 11, which shows that the cumulative sum of R_τ up to round t scales as $\sqrt{t\log t}$ in expectation.

Lemma 11. Consider the setting of Theorem 1. Suppose that we run Algorithm 1 with γ_t, Ψ given by (4), and η_t given by (6). Then, for all $t \geq 1$, we have

$$\mathbb{E}\left[\sum_{\tau=1}^{t} R_\tau\right] = O\left(\sqrt{t\log t}\right).$$

Proof. We first bound $\mathbb{E}[\Phi'_{t+1}(Q_{t+1})]$, which is frequently used throughout the proof. By Lipschitzness, we have $f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star) \geq -LD$. Then it leads to $\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) \geq -LDt$ almost surely. By Lemma 10, we have

$$\mathbb{E}[\Phi_{t+1}(Q_{t+1})] \leq 3DLt + \log t + 1 + D.$$

This leads to

$$\mathbb{E}[\Phi'_{t+1}(Q_{t+1})] \leq \gamma_{t+1}\left(3DLt + \log t + D + 2\right) \leq 3DLt + \log t + D + 2. \tag{17}$$

Now, we bound $\sum_{\tau=1}^{t} R_\tau$. Since $\gamma_\tau \leq 1/(12G\sqrt{\tau})$, we have

$$\mathbb{E}\left[\sum_{\tau=1}^{t} R_\tau\right] \leq \sum_{\tau=1}^{t}\frac{G}{\sqrt{\tau}} + \sum_{\tau=1}^{t}\mathbb{E}\left[\frac{\Psi\left(\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2\right)}{\Phi'_\tau(Q_\tau)}\right].$$

It is clear that $\sum_{\tau=1}^{t} G/\sqrt{\tau} \leq 2G\sqrt{t}$. Moreover, it follows that

$$\begin{aligned}
&\sum_{\tau=1}^{t}\frac{\Psi\left(\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2\right)}{\Phi'_\tau(Q_\tau)} \\
&= 4DL\sum_{\tau=1}^{t}\frac{\sqrt{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2} - \sqrt{\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2}}{\Phi'_\tau(Q_\tau)} \\
&= 4DL\sum_{\tau=1}^{t}\frac{\Phi'_\tau(Q_\tau)}{\sqrt{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2} + \sqrt{\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2}} \\
&\leq 4DL\sum_{\tau=1}^{t}\frac{\Phi'_\tau(Q_\tau)}{\sqrt{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2}} \\
&\leq 4DL\sqrt{t}\sqrt{\sum_{\tau=1}^{t}\frac{\Phi'_\tau(Q_\tau)^2}{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2}} \\
&= 4DL\sqrt{t}\sqrt{\sum_{\tau=2}^{t}\frac{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2 - \sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2}{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2} + 1} \\
&\leq 4DL\sqrt{t}\sqrt{\log\frac{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2}{\Phi'_1(Q_1)^2} + 1} \\
&\leq 4DL\sqrt{t}\sqrt{2\log\frac{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)}{\Phi'_1(Q_1)} + 1}.
\end{aligned}$$

where the second inequality follows from the Cauchy-Schwarz inequality, the third inequality is due to Lemma 31, and the last inequality follows from $\sum_{\tau=1}^t a_\tau^2 \le (\sum_{\tau=1}^t a_\tau)^2$ for any nonnegative sequence $\{a_\tau\}_{\tau=1}^t$. Now, by taking $\mathbb{E}$ on both sides, since $\sqrt{x}$ and $\log x$ are concave functions, it follows that

$$\begin{aligned}
&\sum_{\tau=1}^t \mathbb{E}\left[\frac{\Psi\left(\sum_{s=1}^\tau \Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)}\right] \\
&\le \mathbb{E}\left[4DL\sqrt{t}\sqrt{2\log\frac{\sum_{\tau=1}^t \Phi_\tau'(Q_\tau)}{\Phi_1'(Q_1)}+1}\right] \\
&\le 4DL\sqrt{t}\sqrt{2\log\frac{\sum_{\tau=1}^t \mathbb{E}[\Phi_\tau'(Q_\tau)]}{\Phi_1'(Q_1)}+1} \\
&\le 4DL\sqrt{t}\sqrt{2\log\frac{t(3DLt+\log t+D+2)}{\gamma_1}+1}
\end{aligned}$$

where the second inequality follows from Jensen's inequality, and the last inequality follows from (17). Consequently, we deduce that

$$\mathbb{E}\left[\sum_{\tau=1}^t R_\tau\right] \le 2G\sqrt{t} + 4DL\sqrt{t}\sqrt{2\log\frac{t(3DLt+\log t+D+2)}{\gamma_1}+1}.$$

It can be written as $\mathbb{E}\left[\sum_{\tau=1}^t R_\tau\right] = O\left(\sqrt{t\log t}\right)$. □

6.1 Proof of Theorem 1

Now, we are ready to prove Theorem 1.

Proof. By Lemma 10, since $\Phi_{t+1}(Q_{t+1}) \ge -1$, we have

$$\mathbb{E}[\text{Regret}(t)] = \mathbb{E}\left[\sum_{\tau=1}^t (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star))\right] \le 2DL\sqrt{t} + \log t + 2 + D.$$

Next, to bound $\mathbb{E}[\text{Violation}(t)]$, we observe that $\mathbb{E}\left[\sum_{\tau=1}^t g_\tau(\mathbf{x}_\tau)\right] = \mathbb{E}[Q_{t+1}] + \sum_{\tau=1}^t \mathbb{E}[R_\tau]$. Thus, we first bound $\mathbb{E}[Q_{t+1}]$. As done in the proof of Lemma 11, we have

$$\mathbb{E}[\Phi_{t+1}(Q_{t+1})] \le 3DLt + \log t + 1 + D.$$

By Jensen's inequality, we have $e^{\gamma_{t+1}\mathbb{E}[Q_{t+1}]} - 1 \le \mathbb{E}[e^{\gamma_{t+1}Q_{t+1}} - 1] = \mathbb{E}[\Phi_{t+1}(Q_{t+1})]$. Thus, we have

$$\begin{aligned}
\mathbb{E}[Q_{t+1}] &\le \frac{1}{\gamma_{t+1}}\log\left(3DLt + \log t + 2 + D\right) \\
&\le \left(12G\sqrt{t+1} + 24DL + 1\right)\log\left(3DLt + \log t + 2 + D\right)
\end{aligned}$$

It can be written as

$$\mathbb{E}[Q_{t+1}] = O\left(\sqrt{t}\log t\right).$$

Moreover, by Lemma 11, we have $\mathbb{E}\left[\sum_{\tau=1}^t R_\tau\right] = O(\sqrt{t\log t})$. Consequently, we have

$$\mathbb{E}[\text{Regret}(t)] = O\left(\sqrt{t}\right), \quad \mathbb{E}[\text{Violation}(t)] = O\left(\sqrt{t}\log t\right),$$

as required. $\square$

7 Analysis under Stochastic Constraints: High-Probability Bounds

In this section, we present the analysis of Theorem 2, which establishes high-probability guarantees under convex losses and stochastic constraints. The key challenge is to obtain a high-probability bound on

$$\sum_{\tau=1}^t \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star),$$

which is required to derive a high-probability analogue of Lemma 10. In contrast, in the expected bound setting, the above summation is nonpositive in expectation as a consequence of feasibility, i.e., $\mathbb{E}[g_t(\mathbf{x}^\star)] \leq 0$. Therefore, we must derive a high-probability upper bound on this summation under the sole assumption that $\mathbb{E}[g_t(\mathbf{x}^\star)] \leq 0$. To this end, we introduce the following lemma, which provides an anytime concentration inequality for supermartingales.

Lemma 12. Let $\{X_t\}_{t\geq 1}$ be a random process adapted to a filtration $\{\mathcal{F}_t\}_{t\geq 0}$, satisfying $\mathbb{E}[X_t|\mathcal{F}_{t-1}] \leq 0$ for all $t \geq 1$. Let $\{w_t\}_{t\geq 1}$ be a random process such that w_t is positive and $\mathcal{F}_{t-1}$-measurable for all $t \geq 1$. For some constant G, suppose that $|X_t| \leq G$ almost surely for all $t \geq 1$. Define $\{S_t\}_{t\geq 1}$ such that

$$S_t = \sum_{\tau=1}^t w_\tau X_\tau, \quad \forall t \geq 1.$$

Let $\delta \in (0,1)$. With probability at least $1-\delta$, for all $t \geq 1$,

$$S_t \leq 2G\sqrt{\left(1 + \sum_{\tau=1}^t w_\tau^2\right) \log \frac{(\pi^2/6)(1 + \log_2(1 + \sum_{\tau=1}^t w_\tau^2))^2}{\delta}}.$$

Proof. Since $|X_t| \leq G$, we have $|w_t X_t| \leq G w_t$ almost surely. Fix $\lambda > 0$. By Lemma 33, we have for each $t \geq 1$,

$$\mathbb{E}\left[\exp\left(\lambda w_t X_t\right) | \mathcal{F}_{t-1}\right] \leq \exp\left(\frac{\lambda^2 G^2 w_t^2}{2}\right)$$

Since w_t is $\mathcal{F}_{t-1}$-measurable, it can be rewritten as

$$\mathbb{E}\left[\exp\left(\lambda w_t X_t - \frac{\lambda^2 G^2 w_t^2}{2}\right) | \mathcal{F}_{t-1}\right] \leq 1. \tag{18}$$

Define $\{M_t\}_{t\geq 0}$ such that $M_0 = 1$ and

$$M_t = M_{t-1}\exp\left(\lambda w_t X_t - \frac{\lambda^2 G^2 w_t^2}{2}\right), \quad \forall t \geq 1.$$

Note that $\{M_t\}_{t\geq 0}$ is adapted to $\{\mathcal{F}_t\}_{t\geq 0}$. Moreover, it is a supermartingale, since

$$\mathbb{E}[M_t|\mathcal{F}_{t-1}] \leq M_{t-1}\mathbb{E}\left[\exp\left(\lambda w_t X_t - \frac{\lambda^2 G^2 w_t^2}{2}\right)|\mathcal{F}_{t-1}\right] \leq M_{t-1}.$$

By Lemma 32, it follows that for any $\epsilon > 0$,

$$\mathbb{P}\left(\sup_{t\geq 0} M_t \geq \epsilon\right) \leq \frac{\mathbb{E}[M_0]}{\epsilon} = \frac{1}{\epsilon}.$$

Note that $\mathbb{P}(\sup_{t\geq 1} M_t \geq \epsilon) \leq \mathbb{P}(\sup_{t\geq 0} M_t \geq \epsilon)$. Moreover, $M_t = \exp\left(\lambda S_t - (\lambda^2 G^2/2)\sum_{\tau=1}^t w_\tau^2\right)$ for every $t \geq 1$. By taking $\epsilon = 1/\delta$, it follows that

$$\begin{aligned}
&\mathbb{P}\left(\exists t \geq 1:\ \exp\left(\lambda S_t - \frac{\lambda^2 G^2}{2}\sum_{\tau=1}^t w_\tau^2\right) \geq \frac{1}{\delta}\right) \leq \delta \\
\Leftrightarrow\quad &\mathbb{P}\left(\exists t \geq 1:\ S_t \geq \frac{\lambda G^2}{2}\sum_{\tau=1}^t w_\tau^2 + \frac{1}{\lambda}\log\frac{1}{\delta}\right) \leq \delta.
\end{aligned}$$

Note that this implies that for any $\delta \in (0,1)$,

$$\mathbb{P}\left(\exists t \geq 1:\ S_t \geq \frac{\lambda G^2}{2}\left(1 + \sum_{\tau=1}^t w_\tau^2\right) + \frac{1}{\lambda}\log\frac{1}{\delta}\right) \leq \delta.$$

To conclude the proof, we apply the peeling argument. In particular, let the index set $\mathcal{I}_k$ be defined as for $k \geq 0$,

$$\mathcal{I}_k = \left\{t \geq 1:\ 2^k \leq 1 + \sum_{\tau=1}^t w_\tau^2 < 2^{k+1}\right\}.$$

Furthermore, for each $t \in \mathcal{I}_k$, we have

$$\begin{aligned}
\frac{\lambda G^2}{2}\left(1 + \sum_{\tau=1}^t w_\tau^2\right) + \frac{1}{\lambda}\log\frac{1}{\delta} &< \lambda G^2 2^k + \frac{1}{\lambda}\log\frac{1}{\delta} \\
&= 2G\sqrt{2^k \log\frac{1}{\delta}} \\
&\leq 2G\sqrt{\left(1 + \sum_{\tau=1}^t w_\tau^2\right)\log\frac{1}{\delta}}
\end{aligned}$$

where the equality follows from taking $\lambda = \sqrt{\log(1/\delta)/G^2 2^k}$, and the inequalities follow from the definition of $\mathcal{I}_k$. Then it follows that for any $\delta \in (0,1)$,

$$\mathbb{P}\left(\exists t \in \mathcal{I}_k:\ S_t \geq 2G\sqrt{\left(1 + \sum_{\tau=1}^t w_\tau^2\right)\log\frac{1}{\delta}}\right) \leq \delta.$$

Moreover, given $\delta \in (0,1)$, define $\delta_k = \frac{\delta}{(\pi^2/6)(k+1)^2}$, hence we have $\sum_{k=0}^{\infty} \delta_k = \delta$, as $\sum_{k=0}^{\infty} \frac{1}{(k+1)^2} = \frac{\pi^2}{6}$. By taking $\delta \leftarrow \delta_k$, it follows that

$$\mathbb{P}\left(\exists t \in \mathcal{I}_k :\ S_t \geq 2G\sqrt{\left(1+\sum_{\tau=1}^{t} w_\tau^2\right)\log\frac{1}{\delta_k}}\right) \leq \delta_k$$

$$\Leftrightarrow \quad \mathbb{P}\left(\exists t \in \mathcal{I}_k :\ S_t \geq 2G\sqrt{\left(1+\sum_{\tau=1}^{t} w_\tau^2\right)\log\frac{(\pi^2/6)(k+1)^2}{\delta}}\right) \leq \delta_k.$$

For each $t \in \mathcal{I}_k$, we know that $k \leq \log_2(1+\sum_{\tau=1}^{t} w_\tau^2)$. This implies that

$$\mathbb{P}\left(\exists t \in \mathcal{I}_k :\ S_t \geq 2G\sqrt{\left(1+\sum_{\tau=1}^{t} w_\tau^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t} w_\tau^2))^2}{\delta}}\right) \leq \delta_k.$$

Since $\{t \geq 1\} = \bigcup_{k=0}^{\infty} \mathcal{I}_k$, by taking union bound over $k = 0, 1, 2, \ldots$, we have

$$\mathbb{P}\left(\exists t \geq 1 :\ S_t \geq 2G\sqrt{\left(1+\sum_{\tau=1}^{t} w_\tau^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t} w_\tau^2))^2}{\delta}}\right) \leq \sum_{k=0}^{\infty} \delta_k = \delta.$$

Thus, with probability at least $1-\delta$, for all $t \geq 1$, we have

$$S_t \leq 2G\sqrt{\left(1+\sum_{\tau=1}^{t} w_\tau^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t} w_\tau^2))^2}{\delta}}.$$

$\square$

Note that Lemma 12 suggests that

$$\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) \leq 2G\sqrt{\left(1+\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2))^2}{\delta}}.$$

This observation motivates us to modify the definition of Ψ compared to the expected bound setting. Specifically, since

$$\mathbb{E}\left[\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star)\right] \leq 0$$

in the expected bound setting, the only term that needs to be canceled is

$$2DL\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2},$$

which leads to the choice $\Psi(x) = 4DL\sqrt{x}$. In contrast, under the high-probability setting, the term that must be canceled becomes

$$2DL\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + 2G\sqrt{\left(1+\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2))^2}{\delta}}.$$

This leads to the modified choice of Ψ in (5). While the overall structure of the algorithm remains unchanged, the high-probability setting requires a more sophisticated choice of both Ψ and γ_t. We now proceed to establish high-probability analogues of the previous lemmas under these updated parameter choices.

Lemma 13. Suppose that we run Algorithm 1 with γ_t, Ψ given by (5). Then, for all $t \geq 1$, we have

$$\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t) \leq \Phi_t'(Q_t)\left(g_t(\mathbf{x}_t) - \frac{R_t}{2}\right) + \frac{\gamma_t - \gamma_{t+1}}{e\gamma_{t+1}}. \tag{19}$$

Proof. Note that the left-hand side of (19) can be decomposed as

$$\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t) = \underbrace{\Phi_t(Q_{t+1}) - \Phi_t(Q_t)}_{\text{(I)}} + \underbrace{\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_{t+1})}_{\text{(II)}}. \tag{20}$$

Now, we first bound (I). Recall that $\Phi_t(x) = e^{\gamma_t x} - 1$ and $\Phi_t'(x) = \gamma_t e^{\gamma_t x}$. It follows that

$$\begin{aligned}
\Phi_t(Q_{t+1}) - \Phi_t(Q_t) &= e^{\gamma_t Q_{t+1}} - e^{\gamma_t Q_t} \\
&= e^{\gamma_t Q_t}(e^{\gamma_t(Q_{t+1} - Q_t)} - 1) \\
&= e^{\gamma_t Q_t}(e^{\gamma_t(g_t(\mathbf{x}_t) - R_t)} - 1) \\
&= \frac{1}{\gamma_t}\Phi_t'(Q_t)(e^{\gamma_t(g_t(\mathbf{x}_t) - R_t)} - 1).
\end{aligned}$$

Note that $e^x \leq 1 + x + x^2$ for all $x \leq 1$, $\gamma_t(g_t(\mathbf{x}_t) - R_t) \leq \gamma_t G \leq 1$, and $\Phi_t'(Q_t)/\gamma_t > 0$. Then it follows that

$$\begin{aligned}
&\Phi_t(Q_{t+1}) - \Phi_t(Q_t) \\
&\leq \frac{1}{\gamma_t}\Phi_t'(Q_t)\left(\gamma_t(g_t(\mathbf{x}_t) - R_t) + \gamma_t^2(g_t(\mathbf{x}_t) - R_t)^2\right) \qquad (21) \\
&\leq \Phi_t'(Q_t)\left(g_t(\mathbf{x}_t) - R_t + 3\gamma_t G^2 + 432\gamma_t^3 G^4 + 3\gamma_t\left(\frac{\Psi\left(\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2\right)}{\Phi_t'(Q_t)}\right)^2\right)
\end{aligned}$$

where the second inequality is due to the Cauchy-Schwarz inequality. Since $\gamma_t \leq 1/(12G)$, we have

$$3\gamma_t G^2 + 432\gamma_t^3 G^4 \leq 3\gamma_t G^2 + 432\gamma_t G^4 \cdot \frac{1}{144G^2} = 6\gamma_t G^2. \tag{22}$$

Moreover, we have

$$
\begin{aligned}
&3\gamma_t \frac{\Psi\left(\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2\right)}{\Phi_t'(Q_t)} \\
&= \underbrace{12\gamma_t DL \frac{\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2} - \sqrt{\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2}}{\Phi_t'(Q_t)}}_{\text{(I-a)}} \\
&\quad + \underbrace{12\gamma_t G \frac{\sqrt{\psi\left(\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right)} - \sqrt{\psi\left(\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2\right)}}{\Phi_t'(Q_t)}}_{\text{(I-b)}}
\end{aligned}
\tag{23}
$$

where $\psi(x) = (1+x)\log\frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}$. For (I-a), we have

$$
\begin{aligned}
\text{(I-a)} &= \frac{12\gamma_t DL\Phi_t'(Q_t)}{\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2} + \sqrt{\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2}} \\
&\le 12\gamma_t DL
\end{aligned}
\tag{24}
$$

where the inequality follows from $\Phi_t'(Q_t) \le \sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2} + \sqrt{\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2}$. To bound (I-b), we observe that $\Phi_t'(Q_t) = \sqrt{(1+\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2) - (1+\sum_{\tau=1}^{t-1}\Phi_\tau'(Q_\tau)^2)}$. Then, by Lemma 30, we have

$$
\text{(I-b)} \le 96\gamma_t G\sqrt{\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2))^2}{\delta}}.
$$

To further bound, we observe that $Q_{t+1} \le \sum_{\tau=1}^{t} g_\tau(\mathbf{x}_\tau) \le Gt$ for all $t \ge 1$ and $Q_1 = 0$, implying that $\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2 \le \sum_{\tau=1}^{t}\gamma_\tau^2 e^{2\gamma_\tau G\tau}$. Note that $\gamma_\tau \le 1/(12G\sqrt{\tau})$ and $\gamma_\tau \le 1$. Then it follows that

$$
\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2 \le te^{\sqrt{t}/6}.
$$

Since $\log(1+t) \le t$, it leads to

$$
\log_2\left(1+\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right) = \frac{\log\left(1+te^{\sqrt{t}/6}\right)}{\log 2} \le \frac{\log(1+t)+t/6}{\log 2} \le \frac{t+t/6}{\log 2} = \frac{7t}{6\log 2}.
$$

Hence, it follows that

$$
\log\frac{\pi^2}{6\delta}\left(1+\log_2\left(1+\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right)\right)^2 \le \log\frac{\pi^2}{6\delta}\left(t+\frac{7t}{6\log 2}\right)^2 \le \log\frac{12t^2}{\delta}.
$$

Finally, we deduce that

$$
\text{(I-b)} \le 96\gamma_t G\sqrt{\log\frac{12t^2}{\delta}}. \tag{25}
$$

By substituting (24) and (25) into (23), we have

$$3\gamma_t \frac{\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)}{\Phi'_t(Q_t)} \le \gamma_t \left(12DL + 96G\sqrt{\log\frac{12t^2}{\delta}}\right) \le \frac{1}{2} \tag{26}$$

where the second inequality follows from $\gamma_t \le 1/(24(DL + 8G\sqrt{\log(12t^2/\delta)}))$. Note that Ψ is an increasing function, implying that $\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)$ is non-negative. Then, by applying (22) and (26) to (21), (I) is bounded as

$$\begin{aligned}
\text{(I)} &= \Phi_t(Q_{t+1}) - \Phi_t(Q_t) \\
&\le \Phi'_t(Q_t)\left(g_t(\mathbf{x}_t) - R_t + 6\gamma_t G^2 + \frac{\Psi\left(\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) - \Psi\left(\sum_{\tau=1}^{t-1} \Phi'_\tau(Q_\tau)^2\right)}{2\Phi'_t(Q_t)}\right) \\
&= \Phi'_t(Q_t)\left(g_t(\mathbf{x}_t) - \frac{1}{2}R_t\right).
\end{aligned}$$

Next, we bound (II). Since $\gamma_{t+1} \le \gamma_t$, by Lemma 29, we have

$$\text{(II)} = \Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_{t+1}) \le \frac{\gamma_t - \gamma_{t+1}}{e\gamma_{t+1}}.$$

By applying bounds on (I) and (II) to (20), we conclude the proof. □

The following lemma shows that the cost incurred by using time-varying Lyapunov functions with the modified γ_t defined in (5) is negligible in the sense that its summation scales as $\log t$.

Lemma 14. Consider γ_t given by (5). Then, we have

$$\sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{\gamma_{\tau+1}} \le \frac{3(1+\log t)}{2}.$$

Proof. Recall that $1/\gamma_\tau = \max\{12G\sqrt{\tau}, 24(DL+8G\sqrt{\log(12\tau^2/\delta)}), 1\}$. Note that $\max\{x,y,z\} - \max\{x',y',z\} \le x - x' + y - y'$ for any $x > x'$ and $y > y'$. Moreover, we know that $1/\gamma_\tau \ge 12G\sqrt{\tau}$ and $1/\gamma_\tau \ge 192G\sqrt{\log(12\tau^2/\delta)}$. Then we have

$$\begin{aligned}
\frac{\gamma_\tau - \gamma_{\tau+1}}{\gamma_{\tau+1}} &= \frac{\frac{1}{\gamma_{\tau+1}} - \frac{1}{\gamma_\tau}}{\frac{1}{\gamma_\tau}} \\
&\le \frac{\sqrt{\tau+1} - \sqrt{\tau}}{\sqrt{\tau}} + \frac{\sqrt{\log(12(\tau+1)^2/\delta)} - \sqrt{\log(12\tau^2/\delta)}}{\sqrt{\log(12\tau^2/\delta)}} \\
&\le \frac{1}{2\tau} + \frac{\log(1+1/\tau)}{\log(12\tau^2/\delta)} \\
&\le \frac{1}{2\tau} + \frac{1}{\tau\log(12\tau^2/\delta)} \\
&\le \frac{3}{2\tau}
\end{aligned}$$

where the last two inequality follow from $\log(1+x) \le x$ for any $x \ge 0$ and $\log(12\tau^2/\delta) \ge 1$, respectively. Hence, we have

$$\sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{\gamma_{\tau+1}} \le \sum_{\tau=1}^{t} \frac{3}{2\tau} \le \frac{3(1+\log t)}{2}.$$

□

Lemma 15. Consider the setting of Theorem 2. Suppose that we run Algorithm 1 with γ_t, Ψ given by (5), and η_t given by (6). Then, with probability at least $1-\delta$, for all $t \ge 1$, we have

$$\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \le 2DL\sqrt{t} + \log t + 1 + D + 2G\sqrt{\log \frac{\pi^2}{6\delta}}.$$

Proof. By Lemma 9, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}_\tau) \\
&\le \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + 2DL\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + D.
\end{aligned} \tag{27}$$

Moreover, the first term on the right-hand side can be bounded using Lemma 12. In particular, since $\Phi'_\tau(Q_\tau)$ is $\mathcal{F}_{\tau-1}$-measurable and positive, $g_\tau(\mathbf{x}^\star)$ is $\mathcal{F}_\tau$-measurable satisfying $|g_\tau(\mathbf{x}^\star)| \le G$, and $\mathbb{E}[g_\tau(\mathbf{x}^\star)|\mathcal{F}_{\tau-1}] = \mathbb{E}[g_\tau(\mathbf{x}^\star)] \le 0$, it leads to the following: with probability at least $1-\delta$, for all $t \ge 1$,

$$\begin{aligned}
&\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) g_\tau(\mathbf{x}^\star) \\
&\le 2G\sqrt{\left(1 + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) \log \frac{(\pi^2/6)(1 + \log_2(1 + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2))^2}{\delta}}.
\end{aligned} \tag{28}$$

Moreover, by summing the inequality in Lemma 13 over $\tau = 1, \ldots, t$, since $\Phi_1(Q_1) = 0$, we have

$$\Phi_{t+1}(Q_{t+1}) - \Phi_1(Q_1) \le \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) \left(g_\tau(\mathbf{x}_\tau) - \frac{R_\tau}{2}\right) + \sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}}. \tag{29}$$

By applying (28) and (29) to (27), we have

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \\
&\le 2G\sqrt{\left(1 + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2\right) \log \frac{(\pi^2/6)(1 + \log_2(1 + \sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2))^2}{\delta}} \\
&\quad + 2DL\sqrt{t} + 2DL\sqrt{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2} + D + \sum_{\tau=1}^{t} \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} - \frac{1}{2}\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau) R_\tau.
\end{aligned}$$

Note that

$$\begin{aligned}
\frac{1}{2}\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)R_\tau &\geq \frac{1}{2}\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\cdot\frac{\Psi(\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2)-\Psi(\sum_{s=1}^{\tau-1}\Phi_s'(Q_s)^2)}{\Phi_\tau'(Q_\tau)} \\
&= \frac{1}{2}\Psi\left(\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right)-\frac{1}{2}\Psi\left(0\right) \\
&= 2DL\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2}-2G\sqrt{\log\frac{\pi^2}{6\delta}} \\
&\quad + 2G\sqrt{\left(1+\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2))^2}{\delta}}.
\end{aligned}$$

Thus, we deduce that

$$\begin{aligned}
\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau)-f_\tau(\mathbf{x}^\star))+\Phi_{t+1}(Q_{t+1}) &\leq 2DL\sqrt{t}+\sum_{\tau=1}^{t}\frac{\gamma_\tau-\gamma_{\tau+1}}{e\gamma_{\tau+1}}+D+2G\sqrt{\log\frac{\pi^2}{6\delta}} \\
&\leq 2DL\sqrt{t}+\log t+1+D+2G\sqrt{\log\frac{\pi^2}{6\delta}}
\end{aligned}$$

where the second inequality follows from Lemma 14 and $3/(2e)\leq 1$. This concludes the proof. ∎

Lemma 16. Consider the setting of Theorem 2. Suppose that we run Algorithm 1 with γ_t,Ψ given by (5), and η_t given by (6). Suppose that Lemma 15 holds. Then, for all $t\geq 1$, we have

$$\sum_{\tau=1}^{t}R_\tau = O\left(\sqrt{t}\log(t/\delta)\right).$$

Proof. We first bound $\Phi_{t+1}'(Q_{t+1})$, which is frequently used throughout the proof. By Lipschitzness, we have $f_\tau(\mathbf{x}_\tau)-f_\tau(\mathbf{x}^\star)\geq -LD$. Then it leads to $\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau)-f_\tau(\mathbf{x}^\star))\geq -LDt$. By Lemma 15, we have

$$\Phi_{t+1}(Q_{t+1})\leq 3DLt+\log t+1+D+2G\sqrt{\log\frac{\pi^2}{6\delta}}.$$

This leads to

$$\begin{aligned}
\Phi_{t+1}'(Q_{t+1}) &\leq \gamma_{t+1}\left(3DLt+\log t+2+D+2G\sqrt{\log\frac{\pi^2}{6\delta}}\right) \\
&\leq 3DLt+\log t+2+D+2G\sqrt{\log\frac{\pi^2}{6\delta}}.
\end{aligned}\tag{30}$$

Now, we bound $\sum_{\tau=1}^{t}R_\tau$. Since $\gamma_\tau\leq 1/(12G\sqrt{\tau})$, we have

$$\sum_{\tau=1}^{t}R_\tau\leq\sum_{\tau=1}^{t}\frac{G}{\sqrt{\tau}}+\sum_{\tau=1}^{t}\frac{\Psi\left(\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2\right)-\Psi\left(\sum_{s=1}^{\tau-1}\Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)}.$$

It is clear that $\sum_{\tau=1}^{t} G/\sqrt{\tau} \leq 2G\sqrt{t}$. Moreover, it follows that

$$
\begin{aligned}
&\sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi'_s(Q_s)^2\right)}{\Phi'_\tau(Q_\tau)} \\
&= \underbrace{4DL \sum_{\tau=1}^{t} \frac{\sqrt{\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2} - \sqrt{\sum_{s=1}^{\tau-1} \Phi'_s(Q_s)^2}}{\Phi'_\tau(Q_\tau)}}_{\text{(I)}} \\
&\quad + \underbrace{4G \sum_{\tau=1}^{t} \frac{\sqrt{\psi(\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2)} - \sqrt{\psi(\sum_{s=1}^{\tau-1} \Phi'_s(Q_s)^2)}}{\Phi'_\tau(Q_\tau)}}_{\text{(II)}}.
\end{aligned}
$$

We bound (I) as follows.

$$
\begin{aligned}
\text{(I)} &= 4DL \sum_{\tau=1}^{t} \frac{\Phi'_\tau(Q_\tau)}{\sqrt{\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2} + \sqrt{\sum_{s=1}^{\tau-1} \Phi'_s(Q_s)^2}} \\
&\leq 4DL \sum_{\tau=1}^{t} \frac{\Phi'_\tau(Q_\tau)}{\sqrt{\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2}} \\
&\leq 4DL\sqrt{t} \sqrt{\sum_{\tau=1}^{t} \frac{\Phi'_\tau(Q_\tau)^2}{\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2}} \\
&= 4DL\sqrt{t} \sqrt{\sum_{\tau=2}^{t} \frac{\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2 - \sum_{s=1}^{\tau-1} \Phi'_s(Q_s)^2}{\sum_{s=1}^{\tau} \Phi'_s(Q_s)^2} + 1} \\
&\leq 4DL\sqrt{t} \sqrt{\log \frac{\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2}{\Phi'_1(Q_1)^2} + 1} \\
&\leq 4DL\sqrt{t} \sqrt{\log \frac{t(3DLt + \log t + 2 + D + 2G\sqrt{\log(\pi^2/(6\delta))})^2}{\gamma_1^2} + 1}.
\end{aligned}
$$

where the second inequality follows from the Cauchy-Schwarz inequality, the third inequality is due to Lemma 31, and the last inequality follows from (30), i.e., since $\Phi'_1(Q_1)^2 = \gamma_1^2 \leq 1$,

$$
\begin{aligned}
\sum_{\tau=1}^{t} \Phi'_\tau(Q_\tau)^2 = \sum_{\tau=0}^{t-1} \Phi'_{\tau+1}(Q_{\tau+1})^2 &\leq 1 + (t-1)(3DLt + \log t + 2 + D + 2G\sqrt{\log(\pi^2/(6\delta))})^2 \\
&\leq t(3DLt + \log t + 2 + D + 2G\sqrt{\log(\pi^2/(6\delta))})^2.
\end{aligned}
$$

To bound (II), by Lemma 30,

$$
\begin{aligned}
\text{(II)} &\leq 32G \sum_{\tau=1}^{t} \sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2))^2}{\delta}} \\
&\quad \times \frac{\sqrt{1+\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2}-\sqrt{1+\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2}}{\Phi'_\tau(Q_\tau)} \\
&\leq 32G \sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2))^2}{\delta}} \\
&\quad \times \sum_{\tau=1}^{t} \frac{\sqrt{1+\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2}-\sqrt{1+\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2}}{\Phi'_\tau(Q_\tau)} \\
&\leq 32G\sqrt{t} \sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+t(3DLt+\log t+2+D+2G\sqrt{\log(\pi^2/(6\delta))})^2))^2}{\delta}} \\
&\quad \times \sqrt{\log \frac{t(3DLt+\log t+2+D+2G\sqrt{\log(\pi^2/(6\delta))})^2}{\gamma_1^2}+1}.
\end{aligned}
$$

where the first inequality follows from Lemma 30, and the third inequality follows from the same argument used for bounding (I). Hence, we deduce that

$$
\begin{aligned}
\sum_{\tau=1}^{t} R_\tau &\leq 2G\sqrt{t} + 4DL\sqrt{t}\sqrt{\log \frac{t(3DLt+\log t+2+D+2G\sqrt{\log(\pi^2/(6\delta))})^2}{\gamma_1^2}+1} \\
&\quad + 32G\sqrt{t}\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+t(3DLt+\log t+2+D+2G\sqrt{\log(\pi^2/(6\delta))})^2))^2}{\delta}} \\
&\quad \times \sqrt{\log \frac{t(3DLt+\log t+2+D+2G\sqrt{\log(\pi^2/(6\delta))})^2}{\gamma_1^2}+1}.
\end{aligned}
$$

It can be written as $\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t}\log(t/\delta)\right)$. $\square$

7.1 Proof of Theorem 2

Proof. We assume that Lemma 15 holds, which occurs with probability at least $1-\delta$. Since $\Phi_{t+1}(Q_{t+1}) \geq -1$, we have

$$
\text{Regret}(t) = \sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) \leq 2DL\sqrt{t} + \log t + D + 2G\sqrt{\log \frac{\pi^2}{6\delta}} + 2
$$

Next, to bound Violation(t), we observe that $\sum_{\tau=1}^{t} g_\tau(\mathbf{x}_\tau) = Q_{t+1} + \sum_{\tau=1}^{t} R_\tau$. Thus, we first bound Q_{t+1}. As done in the proof of Lemma 16, we have

$$
\Phi_{t+1}(Q_{t+1}) \leq 3DLt + \log t + 1 + D + 2G\sqrt{\log \frac{\pi^2}{6\delta}}.
$$

By the definition of Φ_{t+1}, since $Q_{t+1} = (1/\gamma_{t+1})\log(\Phi_{t+1}(Q_{t+1})+1)$, we have

$$\begin{aligned}
Q_{t+1} &\leq \frac{1}{\gamma_{t+1}} \log\left(3DLt + \log t + 2 + D + 2G\sqrt{\log\frac{\pi^2}{6\delta}}\right) \\
&\leq \left(12G\sqrt{t+1} + 24\left(DL + 8G\sqrt{\log(12(t+1)^2/\delta)}\right) + 1\right) \\
&\quad \times \log\left(3DLt + \log t + 2 + D + 2G\sqrt{\log\frac{\pi^2}{6\delta}}\right).
\end{aligned}$$

It can be written as

$$Q_{t+1} = O\left(\sqrt{t}\log(t/\delta) + \log^{3/2}(t/\delta)\right)$$

Moreover, by Lemma 16, we have $\sum_{\tau=1}^{t} R_\tau = O(\sqrt{t}\log(t/\delta))$. Consequently, we have

$$\mathrm{Regret}(t) = O\left(\sqrt{t} + \sqrt{\log(1/\delta)}\right), \quad \mathrm{Violation}(t) = O\left(\sqrt{t}\log(t/\delta) + \log^{3/2}(t/\delta)\right).$$

□

8 Analysis under Strongly Convex Losses

In this section, we present the analysis of Theorems 3 and 4. The key difference in the analysis is to derive an analogue of Lemma 9 with an improved guarantee under the modified primal step size defined in (7).

Lemma 17. Consider μ-strongly convex losses for some $\mu > 0$. Suppose that we run Algorithm 1 with η_τ given by (7). Then, for all $t \geq 1$, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)g_\tau(\mathbf{x}_\tau) \\
&\leq \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)g_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2}.
\end{aligned}$$

Proof. For simplicity, let $\hat{f}_\tau = f_\tau + \Phi'_\tau(Q_\tau)g_\tau$ for each $\tau \in [t]$. Note that since $\Phi'_\tau(Q_\tau) > 0$, $\Phi'_\tau(Q_\tau)g_\tau$ is convex, and thus $\hat{f}_\tau$ is μ-strongly convex by Lemma 34. Since projection onto $\mathcal{X}$ is a contraction mapping, we have for any $\mathbf{x}^\star \in \mathcal{X}$,

$$\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2 \leq \|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 + \eta_\tau^2\|\nabla\hat{f}_\tau(\mathbf{x}_\tau)\|^2 - 2\eta_\tau\nabla\hat{f}_\tau(\mathbf{x}_\tau)^\top(\mathbf{x}_\tau - \mathbf{x}^\star).$$

By strong convexity (Lemma 34), we have

$$\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star) + \frac{\mu}{2}\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 \leq \nabla\hat{f}_\tau(\mathbf{x}_\tau)^\top(\mathbf{x}_\tau - \mathbf{x}^\star).$$

Recall that

$$\eta_\tau = \frac{1}{\mu\tau + (2L/D)\sqrt{\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2}}.$$

Then it follows that

$$
\begin{aligned}
&\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star) \\
&\leq \left(\frac{1}{2\eta_\tau} - \frac{\mu}{2}\right)\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \frac{1}{2\eta_\tau}\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2 + \frac{\eta_\tau}{2}\|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2 \\
&= \frac{\mu(\tau-1)}{2}\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \frac{\mu\tau}{2}\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2 \\
&\quad + \frac{L}{D}\sqrt{\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2}(\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2) + \frac{\eta_\tau}{2}\|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2 \\
&\leq \frac{\mu(\tau-1)}{2}\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \frac{\mu\tau}{2}\|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2 \\
&\quad + \frac{L}{D}\sqrt{\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2}(\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2) + \eta_\tau L^2 + \eta_\tau L^2 \Phi_\tau'(Q_\tau)^2
\end{aligned}
$$

where the last inequality follows from that $\|\nabla \hat{f}_\tau(\mathbf{x}_\tau)\|^2 \leq 2\|\nabla f_\tau(\mathbf{x}_\tau)\|^2 + 2\Phi_\tau'(Q_\tau)^2\|\nabla g_\tau(\mathbf{x}_\tau)\|^2 \leq 2L^2(1 + \Phi_\tau'(Q_\tau)^2)$. By summing over $\tau = 1, \ldots, t$, we have

$$
\begin{aligned}
\sum_{\tau=1}^{t}(\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star)) \leq \frac{L}{D}\sum_{\tau=1}^{t}&\sqrt{\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2}(\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2) \\
&+ L^2\sum_{\tau=1}^{t}\eta_\tau + L^2\sum_{\tau=1}^{t}\eta_\tau\Phi_\tau'(Q_\tau)^2.
\end{aligned}
$$

Note that

$$
\sum_{\tau=1}^{t}\sqrt{\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2}(\|\mathbf{x}_\tau - \mathbf{x}^\star\|^2 - \|\mathbf{x}_{\tau+1} - \mathbf{x}^\star\|^2) \leq D^2\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2}.
$$

Moreover, since $\sum_{\tau=1}^{t}(1/\tau) \leq 1 + \log t$, we have

$$
\sum_{\tau=1}^{t}\eta_\tau \leq \sum_{\tau=1}^{t}\frac{1}{\mu\tau} \leq \frac{1+\log t}{\mu}.
$$

Similarly, since $\sum_{\tau=1}^{t} y_\tau/\sqrt{\sum_{s=1}^{\tau} y_s} \leq 2\sqrt{\sum_{\tau=1}^{t} y_\tau}$ for any positive sequence $\{y_\tau\}_{\tau\geq 1}$, we have

$$
\sum_{\tau=1}^{t}\eta_\tau\Phi_\tau'(Q_\tau)^2 \leq \frac{D}{2L}\sum_{\tau=1}^{t}\frac{\Phi_\tau'(Q_\tau)^2}{\sqrt{\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2}} \leq \frac{D}{L}\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2}.
$$

Combining these yields

$$
\sum_{\tau=1}^{t}(\hat{f}_\tau(\mathbf{x}_\tau) - \hat{f}_\tau(\mathbf{x}^\star)) \leq \frac{L^2(1+\log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2}.
$$

Recall that $\hat{f}_\tau = f_\tau + \Phi'_\tau(Q_\tau) g_\tau$. Therefore, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)g_\tau(\mathbf{x}_\tau) \\
&\le \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)g_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2}.
\end{aligned}$$

□

8.1 Expected Bounds

The proofs of the following lemmas are deferred to Appendix A.

Lemma 18. Consider the setting of Theorem 3. Suppose that we run Algorithm 1 with γ_t, Ψ given by (4), and η_t given by (7). Then, for all $t \ge 1$, we have

$$\mathbb{E}\left[\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star))\right] + \mathbb{E}\left[\Phi_{t+1}(Q_{t+1})\right] \le \frac{L^2(1+\log t)}{\mu} + \log t + 1.$$

Lemma 19. Consider the setting of Theorem 3. Suppose that we run Algorithm 1 with γ_t, Ψ given by (4), and η_t given by (7). Then, for all $t \ge 1$, we have

$$\mathbb{E}\left[\sum_{\tau=1}^{t} R_\tau\right] = O\left(\sqrt{t\log t}\right).$$

8.1.1 Proof of Theorem 3

Proof. We closely follow the proof of Theorem 1, where the only difference is that we use Lemma 18 and Lemma 19. By Lemma 18,

$$\mathbb{E}[\text{Regret}(t)] = \mathbb{E}\left[\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star))\right] \le \frac{L^2(1+\log t)}{\mu} + \log t + 2.$$

By applying the same argument in the proof of Theorem 1, we have

$$\begin{aligned}
\mathbb{E}[Q_{t+1}] &\le \frac{1}{\gamma_{t+1}}\log\left(DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 1\right) \\
&\le \left(12G\sqrt{t+1} + 24DL + 1\right)\log\left(DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 1\right)
\end{aligned}$$

which can be written as $\mathbb{E}[Q_{t+1}] = O\left(\sqrt{t}\log t\right)$. Moreover, by Lemma 19, we have $\mathbb{E}\left[\sum_{\tau=1}^{t} R_\tau\right] = O(\sqrt{t\log t})$. Since $\mathbb{E}[\text{Violation}(t)] = \mathbb{E}[Q_{t+1}] + \mathbb{E}\left[\sum_{\tau=1}^{t} R_\tau\right]$, we have

$$\mathbb{E}[\text{Violation}(t)] = O\left(\sqrt{t}\log t\right).$$

□

8.2 High-Probability Bounds

The proofs of the following lemmas are deferred to Appendix A.

Lemma 20. Consider the setting of Theorem 4. Suppose that we run Algorithm 1 with γ_t, Ψ as given by (5), and η_t as given by (7). Then, with probability at least $1-\delta$, for all $t \geq 1$, we have

$$\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \leq \frac{L^2(1+\log t)}{\mu} + \log t + 1 + 2G\sqrt{\log\frac{\pi^2}{6\delta}}.$$

Lemma 21. Consider the setting of Theorem 4. Suppose that we run Algorithm 1 with γ_t, Ψ as given by (5), and η_t as given by (7). Suppose that Lemma 20 holds. Then, for all $t \geq 1$, we have

$$\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t}\log(t/\delta)\right).$$

8.2.1 Proof of Theorem 4

Proof. We closely follow the proof of Theorem 2, where the only difference is that we use Lemma 20 and Lemma 21. We assume that Lemma 20 holds, which occurs with probability at least $1-\delta$. By Lemma 20,

$$\text{Regret}(t) = \sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) \leq \frac{L^2(1+\log t)}{\mu} + \log t + 2 + 2G\sqrt{\log\frac{\pi^2}{6\delta}}.$$

By applying the same argument in the proof of Theorem 2, we have

$$\begin{aligned}
Q_{t+1} &\leq \frac{1}{\gamma_{t+1}}\log\left(DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 2 + 2G\sqrt{\log\frac{\pi^2}{6\delta}}\right) \\
&\leq \left(12G\sqrt{t+1} + 24\left(DL + 8G\sqrt{\log(12(t+1)^2/\delta)}\right) + 1\right) \\
&\quad \times \log\left(DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 2 + 2G\sqrt{\log\frac{\pi^2}{6\delta}}\right)
\end{aligned}$$

which can be written as $Q_{t+1} = O\left(\sqrt{t}\log t + \log^{3/2}(t/\delta)\right)$. Moreover, by Lemma 21, we have $\sum_{\tau=1}^{t} R_\tau = O(\sqrt{t}\log(t/\delta))$. Since $\text{Violation}(t) = Q_{t+1} + \sum_{\tau=1}^{t} R_\tau$, we have

$$\text{Violation}(t) = O\left(\sqrt{t}\log(t/\delta) + \log^{3/2}(t/\delta)\right),$$

as required. □

9 Conclusion

In this paper, we study COCO with stochastic constraints without Slater's condition. We propose a unified primal-dual algorithm with an anytime regularizer and show that it achieves nearly optimal regret and constraint violation guarantees under stochastic constraints. The key insight is that the regularizer stabilizes the dual process by offsetting an accumulated term involving the derivative of an exponential Lyapunov function. We also establish high-probability guarantees and extend the analysis to strongly convex losses and adversarial constraints. Several directions are worth further investigation. One direction is to obtain dynamic regret bounds under stochastic constraints, where dynamic regret is defined with respect to a comparator sequence $\{\mathbf{x}_t^\star\}_{t=1}^T$ satisfying $\{\mathbf{x}_t^\star\}_{t=1}^T \in \arg\min_{\{\mathbf{x}_t\}_{t=1}^T} \sum_{t=1}^T f_t(\mathbf{x}_t)$ s.t. $\mathbb{E}[g_t(\mathbf{x}_t)] \leq 0$ for all $t \in [T]$, while this paper considers a fixed comparator over time. Another direction is to develop projection-free algorithms, as the proposed algorithm requires a projection oracle onto $\mathcal{X}$.

A Deferred Proofs of Section 8

Proof of Lemma 18. We closely follow the proof of Lemma 10, where the only difference is that we use Lemma 17 instead of Lemma 9. Since we take η_t as (7) and γ_t, Ψ as (4), Lemmas 17 and 7 hold. By Lemma 17, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)g_\tau(\mathbf{x}_\tau) \\
&\le \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)g_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2}.
\end{aligned}$$

By Lemma 7,

$$\Phi_{t+1}(Q_{t+1}) - \Phi_1(Q_1) \le \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\left(g_\tau(\mathbf{x}_\tau) - \frac{R_\tau}{2}\right) + \sum_{\tau=1}^{t}\frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}}.$$

Note that

$$\frac{1}{2}\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)R_\tau \ge 2DL\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2}.$$

Then it follows that

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \\
&\le \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)g_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + \sum_{\tau=1}^{t}\frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} \\
&\le \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)g_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + \log t + 1
\end{aligned}$$

where the second inequality follows from Lemma 8 and $1/(2e) \le 1$. By taking $\mathbb{E}$ on both sides, since $\sum_{\tau=1}^{t}\mathbb{E}[\Phi_\tau'(Q_\tau)g_\tau(\mathbf{x}^\star)] \le 0$ by applying the same argument in the proof of Lemma 10, we conclude the proof. □

Proof of Lemma 19. We closely follow the proof of Lemma 11, where the only difference is that we use Lemma 18 instead of Lemma 10. By Lipschitzness and Lemma 18,

$$\mathbb{E}[\Phi_{t+1}'(Q_{t+1})] \le DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 1.$$

Note that

$$\mathbb{E}\left[\sum_{\tau=1}^{t}R_\tau\right] \le \sum_{\tau=1}^{t}\frac{G}{\sqrt{\tau}} + \sum_{\tau=1}^{t}\mathbb{E}\left[\frac{\Psi\left(\sum_{s=1}^{\tau}\Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1}\Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)}\right].$$

It is clear that $\sum_{\tau=1}^t G/\sqrt{\tau} \leq 2G\sqrt{t}$. Moreover, it follows that

$$\begin{aligned}
&\sum_{\tau=1}^t \mathbb{E}\left[\frac{\Psi\left(\sum_{s=1}^{\tau} \Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)}\right] \\
&\leq 4DL\sqrt{t}\sqrt{2\log\frac{\sum_{\tau=1}^t \mathbb{E}[\Phi_\tau'(Q_\tau)]}{\Phi_1'(Q_1)} + 1} \\
&\leq 4DL\sqrt{t}\sqrt{2\log\frac{t(DLt + L^2(1+\log t)/\mu + \log t + 1)}{\Phi_1'(Q_1)} + 1}.
\end{aligned}$$

Finally, we have $\mathbb{E}\left[\sum_{\tau=1}^t R_\tau\right] = O\left(\sqrt{t\log t}\right)$. □

Proof of Lemma 20. We closely follow the proof of Lemma 15, where the only difference is that we use Lemma 17 instead of Lemma 9. Since we take η_t as (7) and γ_t, Ψ as (5), Lemmas 17 and 13 hold. By Lemma 17, we have

$$\begin{aligned}
&\sum_{\tau=1}^t (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^t \Phi_\tau'(Q_\tau) g_\tau(\mathbf{x}_\tau) \\
&\leq \sum_{\tau=1}^t \Phi_\tau'(Q_\tau) g_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^t \Phi_\tau'(Q_\tau)^2}.
\end{aligned}$$

By Lemma 13,

$$\Phi_{t+1}(Q_{t+1}) - \Phi_1(Q_1) \leq \sum_{\tau=1}^t \Phi_\tau'(Q_\tau)\left(g_\tau(\mathbf{x}_\tau) - \frac{R_\tau}{2}\right) + \sum_{\tau=1}^t \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}}.$$

Note that

$$\begin{aligned}
\frac{1}{2}\sum_{\tau=1}^t \Phi_\tau'(Q_\tau) R_\tau \geq\ & 2DL\sqrt{\sum_{\tau=1}^t \Phi_\tau'(Q_\tau)^2} - 2G\sqrt{\log\frac{\pi^2}{6\delta}} \\
&+ 2G\sqrt{\left(1 + \sum_{\tau=1}^t \Phi_\tau'(Q_\tau)^2\right)\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^t \Phi_\tau'(Q_\tau)^2))^2}{\delta}}.
\end{aligned}$$

Then it follows that with probability at least $1-\delta$,

$$\begin{aligned}
\sum_{\tau=1}^t (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) &\leq \frac{L^2(1+\log t)}{\mu} + \sum_{\tau=1}^t \frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} + 2G\sqrt{\log\frac{\pi^2}{6\delta}} \\
&\leq \frac{L^2(1+\log t)}{\mu} + \log t + 1 + 2G\sqrt{\log\frac{\pi^2}{6\delta}}
\end{aligned}$$

where the second inequality follows from Lemma 14 and $3/(2e) \leq 1$. □

Proof of Lemma 21. We closely follow the proof of Lemma 16, where the only difference is that we use Lemma 20 instead of Lemma 15. By Lipschitzness and Lemma 20,

$$\Phi'_{t+1}(Q_{t+1}) \leq DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 2 + 2G\sqrt{\log\frac{\pi^2}{6\delta}}.$$

Note that

$$\sum_{\tau=1}^{t} R_\tau \leq \sum_{\tau=1}^{t} \frac{G}{\sqrt{\tau}} + \sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2\right)}{\Phi'_\tau(Q_\tau)}.$$

It is clear that $\sum_{\tau=1}^{t} G/\sqrt{\tau} \leq 2G\sqrt{t}$. Moreover, by applying the same argument in the proof of Lemma 16,

$$\begin{aligned}
&\sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2\right)}{\Phi'_\tau(Q_\tau)} \\
&\leq \left(4DL + 32G\sqrt{\log\frac{(\pi^2/6)(1+\log_2(1+\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2))^2}{\delta}}\right)\sqrt{t}\sqrt{\log\frac{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2}{\Phi'_1(Q_1)^2}+1}.
\end{aligned}$$

Note that since $\Phi'_{t+1}(Q_{t+1}) = O\left(t + \sqrt{\log(1/\delta)}\right)$, we have $\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2 = O\left(t^3 + t\log(1/\delta)\right)$. Therefore, we have

$$\sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau}\Phi'_s(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1}\Phi'_s(Q_s)^2\right)}{\Phi'_\tau(Q_\tau)} = O\left(\sqrt{t}\log(t/\delta)\right).$$

Consequently, we have $\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t}\log(t/\delta)\right)$. □

B Analysis under Adversarial Constraints

B.1 Convex Losses

Lemma 22. Suppose that we run Algorithm 2 with γ_t, Ψ as given by (4). Then, for all $t \geq 1$, we have

$$\Phi_{t+1}(Q_{t+1}) - \Phi_t(Q_t) \leq \Phi'_t(Q_t)\left(\tilde{g}_t(\mathbf{x}_t) - \frac{R_t}{2}\right) + \frac{\gamma_t - \gamma_{t+1}}{e\gamma_{t+1}}.$$

Proof. Note that $|\tilde{g}_t(\mathbf{x}_t)| \leq |g_t(\mathbf{x}_t)| \leq G$. Therefore, by applying the same argument as in the proof of Lemma 7, we conclude the proof. □

Lemma 23. Consider the setting of Theorem 5. Suppose that we run Algorithm 2 with η_t given by (6). Then, for all $t \geq 1$, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\tilde{g}_\tau(\mathbf{x}_\tau) \\
&\leq \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2} + D.
\end{aligned}$$

Proof. Recall that $\nabla \tilde{g}_t(\mathbf{x}_t) = \nabla g_t(\mathbf{x}_t)$ if $g_t(\mathbf{x}_t) > 0$, and $\mathbf{0}$ otherwise. It follows that $\|\nabla \tilde{g}_t(\mathbf{x}_t)\| \leq L$. Moreover, we have that $\nabla f_t(\mathbf{x}_t) + \Phi_t'(Q_t)\nabla \tilde{g}_t(\mathbf{x}_t)$ is a subgradient of $f_t + \Phi_t'(Q_t)\tilde{g}_t$. Then, by applying the same argument as in Lemma 9, we conclude the proof. $\square$

Lemma 24. Consider the setting of Theorem 5. Suppose that we run Algorithm 2 with γ_t, Ψ given by (4), and η_t given by (6). Then, for all $t \geq 1$, we have

$$\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \leq 2DL\sqrt{t} + \log t + 1 + D.$$

Proof. We closely follow the proof of Lemma 10. By Lemma 23,

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\tilde{g}_\tau(\mathbf{x}_\tau) \\
&\leq \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)^2} + D.
\end{aligned}$$

By Lemma 22,

$$\Phi_{t+1}(Q_{t+1}) - \Phi_1(Q_1) \leq \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\left(\tilde{g}_\tau(\mathbf{x}_\tau) - \frac{R_\tau}{2}\right) + \sum_{\tau=1}^{t}\frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}}.$$

By Lemma 8,

$$\sum_{\tau=1}^{t}\frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} \leq \frac{1+\log t}{2}$$

Combining these results yields

$$\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \leq \sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) + 2DL\sqrt{t} + \log t + 1 + D.$$

Since we have $g_t(\mathbf{x}^\star) \leq 0$ for all $t \geq 1$ in the adversarial constraint setting, we know that $\sum_{\tau=1}^{t}\Phi_\tau'(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) = 0$. This concludes the proof. $\square$

Lemma 25. Consider the setting of Theorem 5. Suppose that we run Algorithm 2 with γ_t, Ψ given by (4), and η_t given by (6). Then, for all $t \geq 1$, we have

$$\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t\log t}\right).$$

Proof. We closely follow the proof of Lemma 11, where the only difference is that we use Lemma 24 instead of Lemma 10. By Lipschitzness and Lemma 24,

$$\Phi_{t+1}'(Q_{t+1}) \leq 3DLt + \log t + 2 + D.$$

Note that

$$\sum_{\tau=1}^{t} R_\tau \le \sum_{\tau=1}^{t} \frac{G}{\sqrt{\tau}} + \sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau} \Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)}.$$

It is clear that $\sum_{\tau=1}^{t} G/\sqrt{\tau} \le 2G\sqrt{t}$. Moreover, it follows that

$$\begin{aligned}
&\sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau} \Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)} \\
&\le 4DL\sqrt{t}\sqrt{2\log \frac{\sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)}{\Phi_1'(Q_1)} + 1} \\
&\le 4DL\sqrt{t}\sqrt{2\log \frac{t(3DLt + \log t + 1 + D)}{\Phi_1'(Q_1)} + 1}.
\end{aligned}$$

Finally, we have $\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t \log t}\right)$. □

B.1.1 Proof of Theorem 5

Proof. Under Algorithm 2, we observe that $\text{Violation}_+(t) = \sum_{\tau=1}^{t} [g_\tau(\mathbf{x}_\tau)]_+ = Q_{t+1} + \sum_{\tau=1}^{t} R_\tau$. By Lemma 24,

$$\text{Regret}(t) = \sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) \le 2DL\sqrt{t} + \log t + 2 + D.$$

By applying the same argument in the proof of Theorem 1, we have

$$\begin{aligned}
Q_{t+1} &\le \frac{1}{\gamma_{t+1}} \log(3DLt + \log t + 1 + D) \\
&\le \left(12G\sqrt{t+1} + 24DL + 1\right) \log(3DLt + \log t + 1 + D)
\end{aligned}$$

which can be written as $Q_{t+1} = O\left(\sqrt{t} \log t\right)$. Moreover, by Lemma 25, we have $\sum_{\tau=1}^{t} R_\tau = O(\sqrt{t \log t})$. Since $\text{Violation}_+(t) = Q_{t+1} + \sum_{\tau=1}^{t} R_\tau$, we have

$$\text{Violation}_+(t) = O\left(\sqrt{t} \log t\right).$$

□

B.2 Strongly Convex Losses

Lemma 26. Consider the setting of Theorem 6. Suppose that we run Algorithm 2 with η_τ given by (7). Then, for all $t \ge 1$, we have

$$\begin{aligned}
&\sum_{\tau=1}^{t} (f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)\tilde{g}_\tau(\mathbf{x}_\tau) \\
&\le \sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) + \frac{L^2(1 + \log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)^2}.
\end{aligned}$$

Proof. Recall that $\nabla \tilde{g}_t(\mathbf{x}_t) = \nabla g_t(\mathbf{x}_t)$ if $g_t(\mathbf{x}_t) > 0$, and $\mathbf{0}$ otherwise. It follows that $\|\nabla \tilde{g}_t(\mathbf{x}_t)\| \leq L$. Moreover, we have that $\nabla f_t(\mathbf{x}_t) + \Phi'_t(Q_t)\nabla \tilde{g}_t(\mathbf{x}_t)$ is a subgradient of $f_t + \Phi'_t(Q_t)\tilde{g}_t$. Then, by applying the same argument as in Lemma 17, we conclude the proof. □

Lemma 27. Consider the setting of Theorem 6. Suppose that we run Algorithm 2 with γ_t, Ψ as given by (4), and η_t as given by (7). Then, for all $t \geq 1$, we have

$$\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \leq \frac{L^2(1+\log t)}{\mu} + \log t + 1.$$

Proof. We closely follow the proof of Lemma 10. By Lemma 26,

$$\begin{aligned}
&\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\tilde{g}_\tau(\mathbf{x}_\tau) \\
&\leq \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + 2DL\sqrt{\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)^2}.
\end{aligned}$$

By Lemma 22,

$$\Phi_{t+1}(Q_{t+1}) - \Phi_1(Q_1) \leq \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\left(\tilde{g}_\tau(\mathbf{x}_\tau) - \frac{R_\tau}{2}\right) + \sum_{\tau=1}^{t}\frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}}.$$

By Lemma 8,

$$\sum_{\tau=1}^{t}\frac{\gamma_\tau - \gamma_{\tau+1}}{e\gamma_{\tau+1}} \leq \frac{1+\log t}{2}$$

Combining these results yields

$$\sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) + \Phi_{t+1}(Q_{t+1}) \leq \sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) + \frac{L^2(1+\log t)}{\mu} + \log t + 1.$$

Since we have $g_t(\mathbf{x}^\star) \leq 0$ for all $t \geq 1$ in the adversarial constraint setting, we know that $\sum_{\tau=1}^{t}\Phi'_\tau(Q_\tau)\tilde{g}_\tau(\mathbf{x}^\star) = 0$. This concludes the proof. □

Lemma 28. Consider the setting of Theorem 6. Suppose that we run Algorithm 2 with γ_t, Ψ as given by (4), and η_t as given by (7). Then, for all $t \geq 1$, we have

$$\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t\log t}\right).$$

Proof. We closely follow the proof of Lemma 19, where the only difference is that we use Lemma 27 instead of Lemma 10. By Lipschitzness and Lemma 27,

$$\Phi'_{t+1}(Q_{t+1}) \leq DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 2.$$

Note that

$$\sum_{\tau=1}^{t} R_\tau \leq \sum_{\tau=1}^{t} \frac{G}{\sqrt{\tau}} + \sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau} \Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)}.$$

It is clear that $\sum_{\tau=1}^{t} G/\sqrt{\tau} \leq 2G\sqrt{t}$. Moreover, it follows that

$$\begin{aligned}
&\sum_{\tau=1}^{t} \frac{\Psi\left(\sum_{s=1}^{\tau} \Phi_s'(Q_s)^2\right) - \Psi\left(\sum_{s=1}^{\tau-1} \Phi_s'(Q_s)^2\right)}{\Phi_\tau'(Q_\tau)} \\
&\leq 4DL\sqrt{t}\sqrt{2\log\frac{\sum_{\tau=1}^{t} \Phi_\tau'(Q_\tau)}{\Phi_1'(Q_1)} + 1} \\
&\leq 4DL\sqrt{t}\sqrt{2\log\frac{t(DLt + L^2(1+\log t)/\mu + \log t + 1)}{\Phi_1'(Q_1)} + 1}.
\end{aligned}$$

Finally, we have $\sum_{\tau=1}^{t} R_\tau = O\left(\sqrt{t\log t}\right)$. □

B.2.1 Proof of Theorem 6

Proof. Under Algorithm 2, we observe that $\text{Violation}_+(t) = \sum_{\tau=1}^{t}[g_\tau(\mathbf{x}_\tau)]_+ = Q_{t+1} + \sum_{\tau=1}^{t} R_\tau$. By Lemma 27,

$$\text{Regret}(t) = \sum_{\tau=1}^{t}(f_\tau(\mathbf{x}_\tau) - f_\tau(\mathbf{x}^\star)) \leq \frac{L^2(1+\log t)}{\mu} + \log t + 2.$$

By applying the same argument in the proof of Theorem 3, we have

$$\begin{aligned}
Q_{t+1} &\leq \frac{1}{\gamma_{t+1}}\log\left(DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 1\right) \\
&\leq \left(12G\sqrt{t+1} + 24DL + 1\right)\log\left(DLt + \frac{L^2(1+\log t)}{\mu} + \log t + 1\right)
\end{aligned}$$

which can be written as $Q_{t+1} = O\left(\sqrt{t}\log t\right)$. Moreover, by Lemma 28, we have $\sum_{\tau=1}^{t} R_\tau = O(\sqrt{t\log t})$. Since $\text{Violation}(t) = Q_{t+1} + \sum_{\tau=1}^{t} R_\tau$, we have

$$\text{Violation}_+(t) = O\left(\sqrt{t}\log t\right).$$

□

C Auxiliary Lemmas

Lemma 29. For $\gamma_1, \gamma_2 > 0$, define $\Phi_1, \Phi_2 : \mathbb{R} \to \mathbb{R}$ as $\Phi_1(x) = e^{\gamma_1 x} - 1$ and $\Phi_2(x) = e^{\gamma_2 x} - 1$ for each $x \in \mathbb{R}$. Suppose that $\gamma_2 \leq \gamma_1$. Then we have $\Phi_2(x) - \Phi_1(x) \leq (\gamma_1 - \gamma_2)/(\gamma_2 e)$ for all $x \in \mathbb{R}$.

Proof. Note that

$$\begin{aligned}
\Phi_2(x) - \Phi_1(x) &= e^{\gamma_2 x} - e^{\gamma_1 x} \\
&= e^{\gamma_2 x}(1 - e^{(\gamma_1 - \gamma_2)x}) \\
&\leq e^{\gamma_2 x}(\gamma_2 - \gamma_1)x \\
&= e^{\gamma_2 x}(-\gamma_2 x) \cdot \frac{\gamma_1 - \gamma_2}{\gamma_2} \\
&\leq \frac{\gamma_1 - \gamma_2}{e\gamma_2}
\end{aligned}$$

where the first inequality follows from $1 - e^{-y} \leq y$ for all $y \in \mathbb{R}$, and the second inequality follows from the fact that $e^{-y}y \leq 1/e$ for all $y \in \mathbb{R}$ and that $(\gamma_1 - \gamma_2)/\gamma_2 \geq 0$. $\square$

Lemma 30. Let $\delta \in (0,1)$. Define $\psi : \mathbb{R}_+ \to \mathbb{R}$ as $\psi(x) = (1+x)\log \frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}$ for each $x \in \mathbb{R}_+$. Suppose that $0 \leq x_1 < x_2$. Then we have

$$\sqrt{\psi(x_2)} - \sqrt{\psi(x_1)} \leq 8\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x_2))^2}{\delta}} \left(\sqrt{1+x_2} - \sqrt{1+x_1}\right).$$

Moreover, we have

$$\frac{\sqrt{\psi(x_2)} - \sqrt{\psi(x_1)}}{\sqrt{x_2 - x_1}} \leq 8\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x_2))^2}{\delta}}.$$

Proof. For $x \in [x_1, x_2]$, we observe that

$$\begin{aligned}
&\frac{d}{dx}\sqrt{\psi(x)} \\
&= \frac{1}{2\sqrt{x+1}}\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}} \\
&\quad + \frac{1}{\log 2 \cdot \sqrt{1+x}(1+\log_2(1+x))\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}}} \\
&\leq \frac{1}{2\sqrt{x+1}}\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}} + \frac{1}{\log 2 \cdot \sqrt{\log(\pi^2/6)}\sqrt{x+1}} \\
&\leq \frac{4}{\sqrt{x+1}}\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x))^2}{\delta}} \\
&\leq \frac{4}{\sqrt{x+1}}\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x_2))^2}{\delta}}.
\end{aligned}$$

Then it follows that

$$\begin{aligned}
\sqrt{\psi(x_2)} - \sqrt{\psi(x_1)} &\leq \int_{x_1}^{x_2} \frac{d}{dx}\sqrt{\psi(x)}dx \\
&\leq 8\sqrt{\log \frac{(\pi^2/6)(1+\log_2(1+x_2))^2}{\delta}} \left(\sqrt{1+x_2} - \sqrt{1+x_1}\right).
\end{aligned}$$

Note that $\sqrt{1+x_2} - \sqrt{1+x_1} \leq \sqrt{x_2 - x_1}$. This leads to the second statement. $\square$

Lemma 31. Let $\{z_t\}_{t\geq 0}$ be an increasing sequence, and let $z_0 > 0$. Then, we have $\sum_{\tau=1}^{t}(z_\tau - z_{\tau-1})/z_\tau \leq \log(z_t/z_0)$ for all $t \geq 1$.

Proof. Note that $\sum_{\tau=1}^{t}(z_\tau - z_{\tau-1})/z_\tau \leq \int_{z_0}^{z_t}(1/x)dx = \log(z_t/z_0)$. □

The following lemma states the maximal inequality for supermartingales (Theorem 3.9 in Lattimore and Szepesvári (2020)), which can be viewed as a Markov-type inequality for the maximum of a supermartingale.

Lemma 32. Let $\{X_t\}_{t\geq 0}$ be a supermartingale with $X_t \geq 0$ almost surely for all $t \geq 0$. Then for every $\epsilon > 0$,

$$\mathbb{P}\left(\sup_{t\geq 0} X_t \geq \epsilon\right) \leq \frac{\mathbb{E}[X_0]}{\epsilon}.$$

The following lemma is a conditional variant of Hoeffding's lemma (Lemma B.7 in Shalev-Shwartz and Ben-David (2014)).

Lemma 33. Let X be a random variable, and let $\mathcal{F}$ be a σ-algebra. Suppose that $\mathbb{E}[X|\mathcal{F}] \leq 0$ and $|X| \leq Y$ almost surely, where Y is a positive $\mathcal{F}$-measurable random variable. Then, for every $\lambda > 0$, almost surely,

$$\mathbb{E}[e^{\lambda X}|\mathcal{F}] \leq e^{\frac{\lambda^2 Y^2}{2}}.$$

Proof. We closely follow the proof of Lemma B.7 in Shalev-Shwartz and Ben-David (2014). Fix $\lambda > 0$. From the proof of Lemma B.7 in Shalev-Shwartz and Ben-David (2014), for any $y > 0$ and $x \in [-y, y]$

$$e^{\lambda x} \leq \frac{(y-x)\,e^{-\lambda y} + (x+y)e^{\lambda y}}{2y}.$$

Since we have $|X| \leq Y$ almost surely and Y is $\mathcal{F}$-measurable, by letting $x \leftarrow X,\ y \leftarrow Y$ and taking $\mathbb{E}[\cdot|\mathcal{F}]$ on both sides, we have

$$\mathbb{E}[e^{\lambda X}|\mathcal{F}] \leq \frac{(Y - \mathbb{E}\left[X|\mathcal{F}\right])\,e^{-\lambda Y} + (\mathbb{E}[X|\mathcal{F}] + Y)e^{\lambda Y}}{2Y} \leq \frac{e^{-\lambda Y} + e^{\lambda Y}}{2}$$

where the second inequality follows from that $\mathbb{E}[X|\mathcal{F}] \leq 0$ and $Y > 0$ imply $\mathbb{E}[X|\mathcal{F}](e^{\lambda Y} - e^{-\lambda Y})/(2Y) \leq 0$. Again, from the proof of Lemma B.7 in Shalev-Shwartz and Ben-David (2014), we know that $(e^{-x} + e^{x})/2 \leq e^{x^2/2}$ for all $x \in \mathbb{R}$. This completes the proof. □

The following lemma states standard properties of strongly convex functions.

Lemma 34. Let $f : \mathbb{R}^d \to (-\infty, \infty]$ be a proper convex function. Suppose that f is μ-strongly convex for some $\mu > 0$. We have for all $\mathbf{x} \in \operatorname{ri}\operatorname{dom}(f), \mathbf{y} \in \operatorname{dom}(f)$, $\nabla f(\mathbf{x}) \in \partial f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x}) + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|_2^2.$$

Moreover, let $g : \mathbb{R}^d \to (-\infty, \infty]$ be a proper convex function. Then, $f + g$ is μ-strongly convex.

Proof. The first statement follows from Theorem 5.24 in Beck (2017) and the fact that a proper convex function f is subdifferentiable in $\operatorname{ri}\operatorname{dom}(f)$ (Theorem 23.4 in Rockafellar (1970)). The second statement follows from Lemma 5.20 in Beck (2017). □

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